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Investing in CO_2 transport infrastructure under uncertainty: A comparison between ships and pipelines

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ABSTRACT

The aim of this study is to assess whether the value of flexibility can influence the investment decision between CO_2 ship and pipeline transport and, therefore, the way the infrastructure develops. For this, the value of a carbon capture and storage project are calculated with the standard net present value (NPV) and with the least-squares Monte Carlo method, which is a real option approach (ROA).

Results of the NPV and ROA show that ships are preferred for small volumes over large distances. For instance, for a design capacity of 2.5 Mt/y, pipelines are preferred for 250 km and ships for 500 km. The ROA shows that the option value to abandon the project and to switch off the CO_2 capture unit temporarily are about 2–4 and 5 times as high for the ship compared to the pipeline configurations, respectively. The option to connect to another storage reservoir has a value of >1000 M€ for the 10 $MtCO_2$/y configurations. Consequently, this option turns the project values positive for the 10 $MtCO_2$/y pipeline and shipping configurations over a distance of 250 and 500 km.

Overall, the value of flexibility did not change the preferred transportation mode from pipeline to ship transport, at least for the considered options to abandon the project, switch off the capture unit temporarily and switch to another storage reservoir. However, under the assumptions made, all 10 $MtCO_2$/y cases were not profitable with the NPV approach, while they were profitable with the ROA.

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1. Introduction

Carbon capture and storage (CCS) has been identified as an essential technology to cost-effectively limit the average global temperature increase to 2 °C above pre-industrial levels (IEA, 2013). With CCS, CO_2 from industrial and energy-combustion related sources is captured from flue gases, and subsequently transported to a geological storage location. CO_2 storage in offshore reservoirs is considered very relevant in Europe because two-thirds of the potential storage locations are situated offshore (EU GeoCapacity, 2008) and offshore CO_2 storage is socially more accepted than onshore CO_2 storage (Mabon et al., 2014; Prangnell, 2013).

There are two modes for offshore CO_2 transport, namely pipeline and ship transport. Both modes are (or have been) evaluated. For instance, the Green Hydrogen project, which is currently on hold, considered shipping for transporting CO_2 from the Netherlands to mature oil fields in Denmark (ZeroCO$_2$, 2015). In contrast, the ROAD and White Rose CCS project, both currently under evaluation, aim to transport CO_2 via pipeline to an offshore reservoir (ROAD CCS, 2015; White Rose, 2015). However, there is limited experience with offshore CO_2 pipeline and ship transport. Nowadays, only one CO_2 pipeline has been installed offshore, namely the 150 km long Snøhvit pipeline, which transports about 0.7 $MtCO_2$ per year (GCCSI, 2013). For shipping, about 3 $MtCO_2$ is annually transported for the (food) industry with small scale ships of 800–1200 m^3 (Decarre et al., 2010; Vermeulen, 2011). These volumes are limited if compared to e.g., the projected amount of 226 $MtCO_2$/y to be captured and stored globally in 2025, in the 2 °C scenario of the International Energy Agency (IEA 2014a).

Both, large scale CO_2 pipeline and ship transport have been studied in the last few years. The configuration and costs of ship transport are reported by, among others, IEA GHG (2004), Aspelund et al. (2006), Decarre et al. (2010), Vermeulen (2011), Nam et al. (2013) and Skagestad et al. (2014). The economics of pipeline transport have been researched by, among others, McCollum and Ogden (2006), McCoy and Rubin (2008), IEA GHG (2002) and Knoope et al. (2013). Comparisons between both transport modes are also available, for instance, by the European Technology Platform for Zero Emission Fossil Fuel Power Plants (ZEP, 2010), Roussanaly et al., (2013a,b, 2014), Jung et al. (2013) and Svensson et al. (2004). These

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Nomenclature

ABEX	abandonment expenditures
C	costs of the component
CAPEX	capital expenditures
CCS	carbon capture and storage
C_{decom_pipe}	decommissioning costs of the pipeline
CF	cash flow
c_n	coefficients estimated by the regression function
DECC	(British) Department of Energy and Climate Change
dW	increment to a standard Wiener process
ETS	emissions trading scheme
EU	European Union
F	regression function
FPSO	floating production, storage and offloading
GBM	geometric Brownian motion
IEA	International Energy Agency
k_e	reverse rate of the electricity price
L	length of the pipeline
LC	levelized costs
L_e	long-run equilibrium electricity price
L_j	Laguerre polynomials of order j
LSMC	least-squares Monte Carlo
MAOP	maximum allowable operation pressure
m_{cap}	captured mass flow
m_{CCS}	CO_2 emitted when CCS is applied
m_{no_CCS}	CO_2 emitted when CCS is not applied
NPV	net present value
O&M	operation and maintenance
OD	outer diameter of the nominal pipe size
OPEX	operational expenditures
$OPEX_{fixed}$	fixed operational expenditures
$OPEX_{var}$	variable operational expenditures
P^c	CO_2 price
Q	value of the CCS project
$\hat{Q}$	Estimated continuation value
r	discount rate
r_f	risk free rate
ROA	real option approach
S	size of component
STL	submerged turret loading system
t	a certain period
T	project duration
WD	water depth
X	value of commodity
y	scale factor
ZEP	European Technology Platform for Zero Emission Fossil Fuel Power Plants
α	constant drift
Δt	time step
σ	constant volatility

Subscripts

c	CO_2
e	electricity
f	coal
g	value of certain simulation path
i	any commodity
o	fuel oil
t	specific period
u	utilization rate

studies show that pipelines realize lower levelized costs than ships for transporting large quantities of CO_2 (>5 $MtCO_2$/y) over short distances (<200 km), while ships realize lower levelized costs over long distances (>1000 km). The break-even distance is estimated at about 700 km for 6.2 $MtCO_2$ per year (IEA GHG, 2004) and it increases with increasing mass flows (IPCC, 2005; Roussanaly et al., 2014). The costs of CO_2 transport are, for instance, 9.3 €/tCO_2 and 14 €/tCO_2 for 2.5 $MtCO_2$/y over 180 km for pipeline and ship transport, respectively (ZEP, 2010). For a distance of 500 km, ships are reported to have levelized costs of 15 €/tCO_2 in comparison with 20 €/tCO_2 for pipelines (ZEP, 2010). At larger distances ship transport is cheaper; the capital expenditures (CAPEX) remain similar while a lower time fraction is spent on mooring, (un)loading and waiting, therefore decreasing the operational expenditures (OPEX) (Nam et al., 2013; Yoo et al., 2013b; ZEP, 2010).

A main difference in the cost structure between ships and pipeline transport is the share of CAPEX and OPEX in the levelized costs. The levelized costs of a pipeline generally consist for about 75–95% of CAPEX, while the costs of ships consist of about 60–80% of OPEX (Bureau et al., 2011; Loeve et al., 2013; Skagestad et al., 2014; ZEP, 2010). Due to the different CAPEX–OPEX structure, it is expected that ships would have a financial advantage compared to pipelines, if the duration of the entire CCS project is shorter than originally anticipated or uncertain altogether (Vermeulen, 2011). The financial advantage of ships could even be enhanced by the residual value of ships (Aspelund et al., 2006) in comparison with significant decommissioning costs for pipelines and platforms (ARUP, 2014; Oil and Gas UK, 2014)[1]. Without incorporating the decommissioning costs, Roussanaly et al. (2014) showed that the break-even distance for an offshore pipeline and ship decreases from about 350 km to 250 km for a mass flow of 5 $MtCO_2$/y, if the project duration decreases from 30 to 10 years.

Another comparative advantage of ship transport is its flexibility. If a storage reservoir is not as large as expected, the ship can route to another storage reservoir (Skagestad et al., 2014; Yoo et al., 2013a). For pipelines, rerouting is more difficult and costly, because an additional section or even a complete new pipeline has to be constructed.

Flexibility has a significant value, which should be taken into account when evaluating the economics of a project (Dixit and Pindyck, 1994). However, the value of flexibility is ignored in the standard net present value (NPV) approach currently used in literature for comparing ships and pipelines. In principle, the NPV approach assumes that volumes, costs, and revenues of a project over the complete project duration are known and adaptation after the investment decision is not possible (or required). In reality, this is not the case, volumes, costs, revenues, and project duration are uncertain and companies will adapt to changing situations. Therefore, a flexible asset is more valuable than an asset which cannot be (easily) adapted to changing situations (Dixit and Pindyck, 1994). A flexible asset gives the option (but not the obligation) to adapt. These options can increase the value of a project, and could lead to a different investment decision than when the value of flexibility is not incorporated.

A method used to evaluate the economic value of options is called the Real Option Approach (ROA). ROA originates from the financial world, but a translation is made to analyze the value of options for investments in tangible assets (Dixit and Pindyck, 1994). Nowadays, ROA has been used to calculate the value of flexibility

[1] About 15 billion € are forecasted to be spent on decommissioning of about 100 platforms, 1000 wells, and 3300 km pipelines on the UK continental shelf in the period 2014–2023 (Oil & Gas UK, 2014).

for different applications ranging from construction management (Ford et al., 2002), investments in existing and new highways (Zhao et al., 2004), to strategies for managing flood risks (Gersonius et al., 2013).

From the available literature, it is not clear whether the explicit inclusion of the value of flexibility will affect the investment decision of ship versus pipeline transport. Therefore, the aim of this study is to investigate the value of flexibility for CO_2 ship and pipeline transport and assess whether this flexibility value influences the investment decision. For this, a whole CCS chain is analyzed consisting of a coal power plant with a CO_2 capture unit, a ship or an offshore pipeline, and storage facilities. For this chain, the project value is calculated with the NPV and with the ROA. Uncertainty is present in many aspects, but the focus in this study is on seven key uncertainties namely, the storage volume of the reservoir, coal price, electricity price, fuel oil price, CO_2 price, and the utilization rate of the coal power plant, which may decrease in the future due to a large penetration of renewables.

2. Real option theory

Real options can be considered as possibilities, and not obligations, to buy, sell or abandon an asset at a given time for a certain price (Dixit and Pindyck, 1994; Hull, 1993). Real options can be exercised when the (economic) situation turns out different than initially expected. There are different kinds of real options, like the option to defer or stage an investment, abandon or expand a project (Fichman et al., 2005). Options improve the potential profit and/or limit losses (Trigeorgis, 1993). For instance, the option to abandon a project can be exercised if the project is unprofitable, thereby limiting the potential losses and increasing the average value of the project. To give a more specific example, De Weck et al. (2004) showed that staged development in the amount of communication satellites reduce the life cycle costs with about 20% compared to the traditional way of designing a large scale satellite at once. Hence, options can change the value of the project and therefore, they should be taken into account when alternatives are compared and investment decisions are taken.

Note that when an option is exercised, the company gives up the possibility of waiting for new information that might affect the desirability or timing of executing the option (Dixit and Pindyck, 1994). For example, after a project is abandoned, it is very costly to restart it when future circumstances improve. Therefore, a certain amount of losses will be tolerated to keep the project alive, before the abandon option is exercised. For more information over the real option theory, see (Dixit and Pindyck, 1994; Hull, 1993). A description of the methodology used in this article to estimate ROA is provided in Section 4.2.

3. CO_2 transportation chains

With CCS, CO_2 is captured from industrial or energy-related sources. After capture, the CO_2 is dried, cleaned and compressed. Capture locations are often not located directly at the coast. Therefore, an onshore pipeline often needs to be constructed to transport the CO_2 to the coastline. At the coastline, the CO_2 can be further transported with an offshore pipeline or a ship to a suitable offshore storage reservoir.

CO_2 pipeline transport is often proposed in the liquid (dense) phase, which is the most efficient for long distances (Knoope et al., 2014). Suitable operation conditions for liquid CO_2 pipeline transport are depicted in Fig. 1. As transporting CO_2 in two phases is undesirable, pumping stations can be installed along the onshore pipeline or before the pipeline goes offshore to compensate for the pressure drop which occur in the pipeline. At the injection site a (unmanned) platform needs to be present. The CO_2 pipeline transport chain is shown in Fig. 2.

Transporting CO_2 with ships on a large scale is often proposed in the liquid form, near the triple point, because the density of the CO_2 is high (Aspelund et al., 2006; ZEP, 2010; Nam et al., 2013; Roussanaly et al., 2013a). However, the CO_2 ships which are currently in operation for the (food) industry use slightly lower pressures and higher temperatures (ZEP, 2010). The possible operation envelop for ship transport stated in the phase diagram in Fig. 1 is based on both indicated conditions. To get the CO_2 into these conditions, the CO_2 has to be liquefied. There are multiple ways to liquefy CO_2. In this study, the compressed CO_2 is liquefied by first depressurizing the CO_2 through expansion valves and, subsequently, separating the liquid CO_2 from the formed gaseous CO_2 by using flash tanks. The gaseous CO_2, which can be 20–40% of the entire CO_2 flow, is recompressed and recycled (Yoo et al., 2013b).

CO_2 capture and liquefaction is a continuous process, while shipping is a batch process (Roussanaly et al., 2013b). Therefore, the liquefied CO_2 is temporarily stored until a ship arrives (ZEP, 2010). The ship will load the liquefied CO_2 on the ship and sail to the injection site. At the injection site, the ship unloads the CO_2 as quickly as possible to a temporary storage vessel[2]. The CO_2 is not injected directly into the reservoir because the injection process prefers to have a continuous and stable CO_2 flow (Roussanaly et al., 2014). Moreover, the storage reservoir cannot handle the low pressure and low temperature of the liquefied CO_2. Consequently, the CO_2 is first pumped and subsequently heated before the CO_2 is offloaded to the injection wells (Roussanaly et al., 2014; Vermeulen, 2011). The CO_2 ship transport chain is shown in Fig. 2.

4. Method

In this study, the decision is analyzed to invest in CO_2 pipeline or ship transport. Although this article focuses on the transportation chain, the capture unit and the platform are incorporated to be able to analyze the costs of the CCS chain and compare them to the CO_2 price. This study excludes the costs of the industrial facility or power plant, well drilling and abandoning, and monitoring of the storage reservoir, see Fig. 2.

4.1. Net present value approach

An investment decision in the NPV approach is based on whether the project generates a positive NPV. Required inputs for the NPV approach are the estimated costs and revenues or benefits during the project duration. With respect to CCS, benefits will be present in the form of less CO_2 emission allowances that have to be bought. Besides calculating the NPV, the levelized costs (LC) are calculated, reflecting the average costs of capturing, transporting and storing 1 t of CO_2 throughout the project lifetime.

The NPV and the LC are calculated with Eqs. (1) and (3), respectively. Summation formulas are used because the utilization rate is expected to decrease and the CO_2 price is expected to increase over time. Consequently, the operational expenditures and benefits are not constant over time. In Table 1, the different CAPEX, OPEX and abandonment expenditures (ABEX), which are included in this

[2] Some literature sources state that a floating vessel is not needed and that the CO_2 can be (after been heated and pumped) directly injected from the ship into the reservoir (Aspelund et al., 2006; Decarre et al., 2010; Vermeulen, 2011; ZEP, 2010). However, transient operation could lead to depressurization of the well. This can result in associated problems like thermal stress and the formation of hydrates in the well (De Koeijer et al., 2014). Therefore, to avoid these problems it is conservatively assumed that an offshore storage vessel is required.

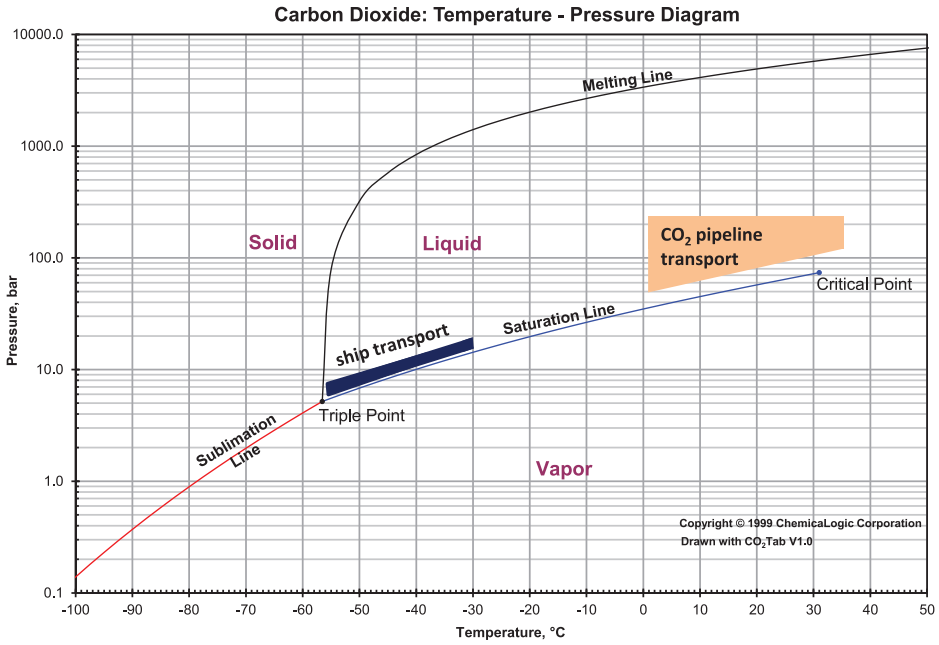


Fig. 1. Phase diagram for pure CO_2 (adapted from ChemicaLogic, 1999) with typical operation envelopes for CO_2 pipeline (based on DNV, 2010; ZEP, 2010) and ship transport (based on ZEP, 2010).

Table 1
Different CAPEX, OPEX and ABEX included in this study for the shipping and pipeline solution.

	CAPEX	OPEX	ABEX[a]
Pipeline	Capture and compression Pumping station Offshore pipeline Platform	Fixed operation and maintenance (O&M) costs capture plant Variable capture costs Electricity (pumps) O&M costs pipeline O&M costs platform	Capture and compression plant Pumping station Offshore pipeline Platform
Ship	Capture and compression Liquefaction unit Temporal storage (Un)loading equipment Ship including cargo pumps Floating vessel with conditioning equipment	Fixed O&M costs capture plant Variable capture costs Electricity (liquefaction) O&M costs ship Fuel oil for ship O&M costs floating vessel and conditioning equipment Conditioning energy	Capture and compression plant Liquefaction unit Temporal storage (Un)loading equipment Ship Floating vessel with conditioning equipment

[a] ABEX can be negative if the residual value is higher than the costs of decommissioning.

study, are given for ship and pipeline transport. The transportation alternative with the highest NPV (or lowest LC) is selected.

A sensitivity analysis is performed to assess the impact on the NPV of higher and lower initial utilization rate, coal price, initial CO_2 price, electricity price, or fuel oil price. The included uncertainty ranges are given in Table 2.

$$\mathrm{NPV} = \sum_{t=0}^{T+0.25} \left[\left(P_t^c \times (m_{\mathrm{no_CCS}_t} - m_{\mathrm{CCS}_t}) - \mathrm{OPEX}_t - \mathrm{CAPEX}_t - \mathrm{ABEX}_t \right) \times e^{-rt} \right] \quad (1)$$

$$P_t^c = P_0^c \times (1 + \alpha_c)^t \quad (2)$$

$$\mathrm{LC} = \sum_{t=0}^{T+0.25} \left(\frac{\mathrm{OPEX}_t + \mathrm{CAPEX}_t + \mathrm{ABEX}_t}{m_{\mathrm{cap}_t}} \times e^{-rt} \right) \quad (3)$$

where NPV is the net present value; T is the project duration (y) and the abandonment expenditures are realized in the period $T+0.25$; $P^c{}_t$ is the CO_2 price in period t (€/tCO_2); $m_{\mathrm{no_CCS}_t}$ is the CO_2 emitted when CCS is not applied in period t (tCO_2/y); m_{CCS_t} is the CO_2 emitted when applying CCS in period t (tCO_2/y); OPEX_t, CAPEX_t and ABEX_t refer to the operational, capital and abandonment expenditures in period t, respectively (€/y); r is the discount rate; $P^c{}_0$ is the CO_2 price in year 0 (€/tCO_2); α_c is the growth rate or constant drift of the CO_2 price (%); LC are the levelized costs (€/tCO_2); and m_{cap_t} is the amount of CO_2 captured in period t (tCO_2/y).

4.2. Real option approach

In this study, the real option approach is used to analyze the value of flexibility. Three different flexibility options are analyzed in this study. First, there is the option to switch to another storage reservoir when the original reservoir is full. The consequences of this for the two transportation solutions are different. For the pipeline solution, the pipeline has to be extended from the old to the new storage location, a new platform has to be constructed, and the old platform has to be decommissioned. For the shipping

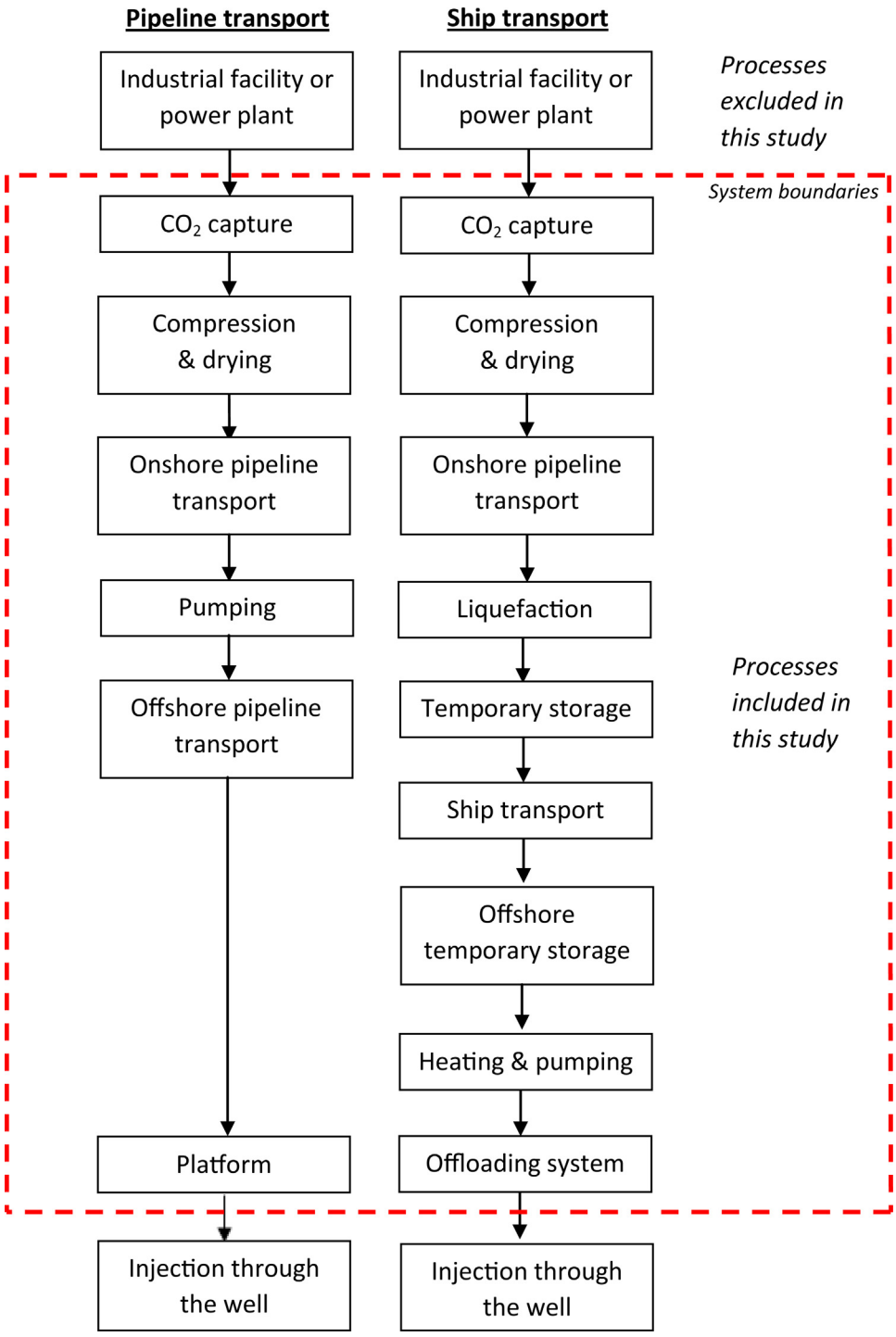


Fig. 2. Schematic overview of the transport chains. On the left side, the transportation chain for the offshore pipeline solution is given and on the right side the transportation chain for ship transport is pictured. The processes within the dotted system boundaries are included in this study.

solution, the floating vessel has to be moved and the ship has to sail to a different location.

Second, the CO_2 capture unit can be temporarily switched off if the CO_2 price does not compensate the variable capture, transport and storage costs. If this happens, the fixed OPEX still have to be paid (e.g., permanent staff, insurance), but the unit can be turned on if the CO_2 price increases (or the variable OPEX decreases) in the next period.

Third, the CCS project can be abandoned during the entire lifetime of the project. In this case, the ship is assumed to be sold for a residual value, as it can also be used for different purposes. The other infrastructure components (like the capture and compressor unit, liquefaction unit, pumps, pipelines, and platforms) for the ship as well as for the pipeline solution have to be decommissioned. In this case, starting up the CCS project again would be very costly and is assumed to be unfeasible.

The value of the different options are calculated for each option separately and for all options combined.

4.2.1. Least-squares Monte Carlo approach

In this study, the different options cannot only be exercised on a fixed date, but also during the entire project duration.

Furthermore, multiple uncertainties are assessed and many of them have a stochastic behavior. Simulation techniques are the best method for analyzing problems with these characteristics (Abdel Sabour and Dimitrakopoulos, 2011; Longstaff and Schwartz, 2001). In this study, the least-squares Monte Carlo (LSMC) approach, developed by Longstaff and Schwartz (2001), is used to value the different options and compare the pipeline and ship solutions.

In this study, different uncertainties are included and modeled stochastically. This means that expectations of future values are based on the current value, but elements of randomization are present in predicting the future. Two different stochastic relations are used in this study. First, the Geometric Brownian Motion (GBM), given in Eq. (4), is used for the load factor, CO_2, coal and fuel oil prices. Coal and oil prices are assumed to follow a GBM, because the mean reversion rate is rather slow (Pindyck, 1999). Also the CO_2 price is assumed to follow a GBM, which is consistent with (Abadie and Chamorro, 2008; Fuss et al., 2008; Zhang et al., 2014; Zhu and Fan, 2011). Furthermore, the utilization rate of the coal power plant is also assumed to follow a GBM. Currently, (most) coal power plants operate on base load. However, if more renewables penetrate the power system over time, then base load plants have to operate with more flexibility. Hence, the utilization rate would decrease over time (i.e., a negative drift). In addition, there is uncertainty in the utilization rate of the coal power plant due to variation in the generated electricity by renewables, variation in the allocation of (different types of) power plants, etc.

Second, the electricity price is described with a GBM with mean reversion, see Eq. (5), implying that the electricity price has a tendency to merge to a long term mean level (Abadie and Chamorro, 2008). Besides the two different GBM models, the uncertainty in the storage reservoir is modeled with a normal distribution, to make outcomes around the mean most likely, but also include that considerably lower or higher values may occur.

Note that different variables can be correlated. For instance, it is likely that a high coal price would result in an increase in the electricity price. Therefore, correlation coefficients have been defined.

$$dX_i = \alpha_i x_i dt + \sigma_i x_i \, dW_i \tag{4}$$

$$dX_e = k_e (L_e - X_e) \, dt + \sigma_e X_e dW_e \tag{5}$$

where X_i is the value of i and i can reflect coal ($_f$), CO_2 ($_c$), electricity ($_e$), fuel oil ($_o$) or utilization rate ($_u$); α_i is the constant drift or growth rate of i (%); σ_i is the constant volatility or standard deviation of i (%); dW_i is the increment to a standard Wiener process, which is normally distributed with a mean of zero and a variance of dt; k_e is reverse rate of the electricity price; and L_e is the long-run equilibrium electricity price.

Based on the stochastic relations, 10,000 runs are generated with a Monte Carlo analysis. For each run and each period, the cash flows for the pipeline and ship solution are calculated, see Eq. (6). Thereby, a distinction is made between fixed and variable OPEX, because the variable OPEX can be avoided if no CO_2 is captured, transported and stored when the option is present to temporarily switch off the capture unit. Following Fleten and Näsäkkälä (2010) and Abadie (2015), the ramping down and starting up costs are assumed to have only a minor effect on the overall valuation, and they are therefore ignored in this study. This implies that it is best to switch off the capture unit, if the CO_2 price does not compensate

Table 2
Economic parameters for ship and pipeline transport used in this study.

Parameter	Unit	Value	Comment/source
Both ship and pipeline			
Design lifetime CCS project	Year	25	[a]
Time step (Δt)	Year	0.25	Own assumption
Construction and decommissioning period	Year	0.25	[b]
Risk free rate (r_f)	%	5	[c]
Discount rate (r)	%	10	[d]
Uncertainty range discount rate for sensitivity analysis	%	50–150% (5–15%)	[d]
Initial CO_2 price	€/tCO_2	35	[e]
Volatility CO_2 price	%	47	[e]
Drift CO_2 price	%	3.1	[e]
Uncertainty range initial CO_2 price for sensitivity analysis	% (€/tCO_2)	50–200% (17.5–70)	[f]
Initial coal price	€/GJ	2.5	[g]
Volatility coal price	%	8	[g]
Drift coal price	%	0	[g]
Uncertainty range coal price for sensitivity analysis	% (€/GJ)	80–120% (2.0–3.0)	[f]
Initial utilization rate coal power plant	%	85	[h]
Volatility utilization rate	%	5	[i]
Drift utilization rate	%	−2	[i]
Uncertainty range utilization rate for sensitivity analysis	% (%)	75–106% (64–90%)	[f]
Initial cost of electricity	€/MW h	60	[j]
Volatility electricity price[l]	%	50	Abadie and Chamorro (2008)
Reverse rate of the electricity price[l]		0.96	Abadie and Chamorro (2008)
Long-run equilibrium electricity price	€/MW h	70	[j]
Uncertainty range electricity price for sensitivity analysis	% (€/MW h)	70–260% (42–156)	[k]
Ship			
Price of fuel oil[l]	€/t_{fuel}	500	INSEE (2014)
Volatility fuel oil price	%	18	[l]
Drift fuel oil price	%	0	[l]
Uncertainty range fuel oil price for sensitivity analysis	% (€/t_{fuel})	60–150% (300–750)	[f]
Ship speed (independent of size)	knots	16.5	Roussanaly et al. (2013a)
Harbor fee	€/tCO_2	1	Roussanaly et al. (2013a)
Availability ship[m]	Days	350	Roussanaly et al. (2013b)
Loading time	Hours	12	ZEP (2010)
Unloading time to buffer storage	Hours	12	ZEP (2010)
Liquefaction energy[n]	kW h/tCO_2	39	Yoo et al. (2013b)
Cooling water use for liquefaction	m^3/tCO_2	3.38	ZEP (2010)
Costs of cooling water	€/m^3	0.14	ZEP (2010)
Conditioning energy[o]	t_{fuel}/tCO_2	0.66	Apeland et al. (2011b)

Table 2 (*Continued*)

Parameter	Unit	Value	Comment/source
Pipeline			
Pumping[p]	kW h/tCO_2 per MPa	0.44	Knoope et al. (2014)
Fixed OPEX costs pipeline	% of CAPEX	1.5	Knoope et al. (2014)
Fixed OPEX costs pumping station	% of CAPEX	4.0	Knoope et al. (2014)
Fixed OPEX platform costs	% of CAPEX	5.0	Van den Broek et al. (2010)

[a] In literature, design lifetimes of pipelines vary between 20 and 50 years (Chandel et al., 2010; ElementEnergy, 2010; Knoope et al., 2014; McCollum and Ogden, 2006; McCoy and Rubin, 2008; Van den Broek et al., 2010; ZEP, 2010), while lifetime of CO_2 ships vary between 15 and 40 years (Apeland et al., 2011a,b; Aspelund et al., 2006; Decarre et al., 2010; Nam et al., 2013; Roussanaly et al., 2013a; ZEP, 2010). In this study, the lifetime of ships and pipelines are assumed to be the same for the sake of simplicity and comparison. Note that the actual lifetime of the project may be shorter than the design lifetime of 25 years because the CCS project could be abandoned earlier.

[b] For simplicity reasons, the construction (and abandonment) period is assumed to be equal to Δt, which is 3 months. However, it is acknowledge that, in reality, planning and construction (or abandonment) activities could take several years.

[c] The risk free rate is within the range of 4–5% used for CCS projects in literature (Abadie and Chamorro, 2008; Ho and Liu, 2002; Sarkis and Tamarkin, 2005; Zhang et al., 2014; Zhu and Fan, 2011).

[d] The continuously compounded discount rate is set at 10%, which equals the risk free rate plus a market risk premium of 5% to incorporate that a higher return is required for investments, which are not risk free. This is comparable to the risk premium of 4.5–5.5% estimated by several models (Koller et al., 2010). Furthermore, the resulting discount range is within the discount rate used for ship transport of 5–10% (Apeland et al., 2011b; Aspelund et al., 2006; Decarre et al., 2010; Nam et al., 2013; Roussanaly et al., 2013a; ZEP, 2010) and for pipelines of 5–15% (Chandel et al., 2010; ElementEnergy, 2010; Knoope et al., 2014; McCollum and Ogden, 2006; McCoy and Rubin, 2008; Van den Broek et al., 2010; ZEP, 2010). For the sensitivity analysis, a discount range of 5–15% is used. In this study, the risk premium and discount rate for ships and pipelines are assumed to be the same for the sake of simplicity and comparison.

[e] In 2014, the CO_2 price was about 5 €/tCO_2 in the European Union Emissions Trading Scheme (EU ETS). This price is too low to cover the variable costs of capture, transport and storage. It is estimated by ZEP that a price of at least about 35 €/tCO_2 is needed to make CCS commercially viable, which is in accordance with the expected EU ETS price of 2025 (ZEP, 2011b). The volatility and drift are based on the CO_2 allowances traded in the context of the second phase (Dec-2008 and Dec-2012) of the EU ETS (Abadie and Chamorro, 2008).

[f] The uncertainty ranges for the sensitivity analysis of the coal and fuel oil price for the NPV approach are based on the 5–95% confidence interval of the average value of the Monte Carlo simulation runs. The uncertainty ranges of the CO_2 price and utilization rate are based on the 5–95% confidence interval of the projected values in year 25, which are subsequently converted to an estimated initial value by using the corresponding drifts. To avoid that the utilization rate becomes higher than 100%, the upper value of the utilization rate is based on the average utilization rate of a coal power plant with CCS estimated by (Mott MacDonald, 2010).

[g] The coal price is based on the average price of imported coal (from overseas) to the Netherlands in the period 2003–2013 (Statline, 2014). The historical volatility of the coal price is estimated to be 7–9% and the growth rate is close to zero (Pindyck, 1999). In this study, the average of the volatility range is used.

[h] The utilization rate is similar to the 7500 h assumed by ZEP (2011a). It is assumed that the operation hours of the coal power plant are spread evenly throughout the year.

[i] The volatility for the utilization rate is based on quarterly utilization rates of coal power plants from 2000 to 2012 in the United States (EIA, 2015a,b). It is estimated that the utilization rate of base load plants (nuclear and geothermal plants with CCS) would drop to about 60%, with a penetration rate of 80% renewables (Brouwer, in press). A slightly lower utilization rate of 50% is found for coal power plants with CCS in Europe, where 80% of the electricity is generated by renewables (Bertsch et al., 2012). The International Energy Agency (IEA) projects average utilization rates for coal power plants with CCS of 69–85% in 2020–2040 and of 53–85% in 2050 (IEA, 2014a). In this study, the average utilization rate is assumed to decline over time from 85% to 50% in 25 years, i.e., a growth rate of −2% per year.

[j] The electricity price is based on the average price of large industrial consumers (>20 GW h/y) in the EU-28 (weighted according to their total electricity consumption) in the period 2007–2012 (Eurostat, 2014). The electricity price is very dynamic, but there is a difference in the dynamics of the short (hour/day) and long term (month/year). In this study, the long term dynamics are considered relevant, and the drift and volatility are based on monthly data (Abadie and Chamorro, 2008). The mean reverting long term electricity price is assumed to increase linearly from the current level of 60 €/MW h to 70 €/MW h in year 25. This increase is comparable to the average electricity price increase foreseen in Europe, China and the U.S. from 2012 to 2040 (IEA, 2014b).

[k] The electricity price range is based on the uncertainty range for the electricity costs with CO_2 capture indicated by ZEP (2011a), of 42–157 €/MW h for a *n*th kind of a plant.

[l] The costs are based on heavy fuel oil with 1% sulfur. These costs are within the range mentioned in literature, which is 430–790 €/t_{fuel} (Apeland et al., 2011b; Roussanaly et al., 2013a; ZEP, 2010). The price of fuel oil is assumed to be fully positively correlated to the oil price. The historical volatility of oil prices is estimated to be 16–21% and the growth rate is close to zero (Pindyck, 1999). In this study, the average of the volatility range is used.

[m] 15 days per year are used for maintenance. It is assumed that this take place during off-peak periods.

[n] Different energy consumption values for liquefaction are quoted in literature. However, the majority of studies starts with CO_2 at atmospheric pressures, while the configuration in this study starts with CO_2 of 10 MPa, which is similar to the inlet conditions of Yoo et al. (2013b), ZEP (2010) and IEA GHG (2004). IEA GHG (2004) mentions an energy consumption of 14.4 kW h/tCO_2, but the formed gaseous CO_2 is vented instead of recompressed. Yoo et al. (2013b) indicate an energy consumption of 17 kW h/tCO_2 for a multi-stage compression configuration. They indicate that a multi-stage configuration is 44% more efficient than a one-stage configuration. This implies that with a one-stage compressor configuration, the energy consumption would be 39 kW h/tCO_2. Unfortunately, the CAPEX of the different configurations are not indicated by Yoo et al. (2013b). The CAPEX (but not the energy consumption of) a one-stage compressor configuration is given in ZEP (2010). Hence, the CAPEX of ZEP and energy consumption of Yoo et al., for a one stage compressor configuration are used. The required energy for liquefaction is assumed to be purchased from the grid (ZEP, 2010).

[o] The conditioning energy for pumping and heating CO_2 from −50 °C and 0.7 MPa to 0 °C and 7 MPa is estimated on 3 kW h/tCO_2 (Apeland et al., 2011b). This electricity is consumed by pumps. The electricity is generated by the engine of the vessel, which leads to an additional fuel oil consumption of 0.220 kg_{fuel}/kW h and consequently to a conditioning energy of 0.66 kg_{fuel}/tCO_2 (Apeland et al., 2011b). The CO_2 will be warmed up by heat exchangers, which will use waste heat of the engine and seawater.

[p] The pumping energy is based on an efficiency of 75% and on a CO_2 density of 850 kg/m^3, which is related to CO_2 with of 10 MPa and 20 °C.

the variable OPEX. The option value of switching off the capture unit temporarily can be found by comparing the sum of the discounted cash flows with and without the option to switch off the CO_2 capture unit.

For the abandoning option, the question is whether it is better to abandon or continue with the project in each period *t*. Note that in the final period, the project has to be abandoned, see Eq. (7). Prior to the final period, the value of the project is the current cash flow plus the sum of the future cash flow(s), see Eq. (8). The future cash flows are estimated by using the simulation results from the Monte Carlo analysis and a least square regression (Longstaff and Schwartz, 2001). More specifically, the ex post realized cash flows of all individual simulation paths from continuing at period $t+1$ are regressed by using the values of the stochastic variables as independent parameters at period *t*. After the future cash flows are estimated for each period, it is evaluated, along each path for each discrete time period from time zero till the end of the project duration, if abandoning is more cost-effective than continuing (Abdel Sabour and Poulin, 2006; Longstaff and Schwartz, 2001; Zhu and Fan, 2011). If the sum of the estimated future cash flows is lower than the ABEX of the overall project, than it is better to abandon the project, see Eq. (9). Subsequently, the value of the project is found by discounting all resulting cash flows with the risk free rate to time zero and calculating the average of the project value from all paths.

For estimating the future cash flows for the abandoning option, only simulation results are used for the regression exercise with a cash flow of zero or lower. This gives a more accurate

approximation of the value of continuing the project and increases process efficiency[3]. Note that for each time period the actual discounted cash flows are used rather than the conditional expected value estimated in the previous period, because this would lead to an upward bias in the option value (Longstaff and Schwartz, 2001).

In this study, the regression functions are based on the first three weighted Laguerre polynomials, see Eqs. (10)–(12), as these provide more accurate results than other polynomial relations (Areal et al., 2008; Moreno and Navas, 2003). Besides the weighted Laguerre polynomials, also cross-products of the variables are included to improve the fit of the regression function. Initial analysis shows that the electricity price has very little prediction power, most probably due to the strong reverse rate in the GBM with mean reversion. Hence, the electricity price is not included in the regression function. Overall, this leads to the regression function shown in Eq. (13).

$$\mathrm{CF}_t = \mathrm{Max}\left(P^c_t \times (m_{no_CCS_t} - m_{CCS_t}) - \mathrm{OPEX}_{var_t} \times m_{cap_t}; 0\right) - \mathrm{OPEX}_{fixed_t} - \mathrm{CAPEX}_t \quad (6)$$

$$\mathrm{CF}_{g,T} = -\mathrm{ABEX}_T \quad (7)$$

$$Q(g,t) = \mathrm{CF}_{g,t} + \mathrm{CF}_{g,t+1} * e^{-r_f \Delta t} \quad (8)$$

$$Q(g,t) = \begin{cases} \mathrm{ABEX}_t & \text{if} \quad \hat{Q}(g,t) < \mathrm{ABEX}_t \\ \hat{Q}(g,t) & \text{if} \quad \hat{Q}(g,t) > \mathrm{ABEX}_t \end{cases} \quad (9)$$

$$L_0(X_i) = \exp\left(\frac{-X_i}{2}\right) \quad (10)$$

$$L_1(X_i) = \exp\left(\frac{-X_i}{2}\right)(1 - X_i) \quad (11)$$

$$L_2(X_i) = \exp\left(\frac{-X_i}{2}\right)\left(1 - 2X_i + \frac{{X_i}^2}{2}\right) \quad (12)$$

$$F_t = \sum_{j=0}^{2}\sum_{i}^{I}\left[a_{ij}L_j(X_i)\right] + c_0 + c_1X_cX_f + c_2X_cX_u + c_3X_cX_o + c_4X_fX_o + c_5X_fX_u + c_6X_uX_o \quad (13)$$

where CF_t is the cash flows in period t; ${P^c}_t$ is the CO_2 price in period t (€/tCO_2); $m_{no_CCS_t}$ is the CO_2 emitted when CCS is not applied in period t (tCO_2/y); m_{CCS_t} is the CO_2 emitted when applying CCS in period t (tCO_2/y); OPEX_{var_t} are the variable operational expenditures per tonne of CO_2 captured in period t (€/tCO_2); m_{cap_t} is the captured mass flow (tCO_2/y); OPEX_{fixed_t}, CAPEX_t and ABEX_t refer to the fixed operational, capital and abandonment expenditures in period t, respectively (€/y); $\mathrm{CF}_{g,t}$ is the cash flow of simulation path g in time period t; T is the project duration (y); Q (g,t) is the value of the CCS project of path g in period t; r_f is the risk free rate; Δt is the time step; $\hat{Q}(g,t)$ is the estimated continuation value of path g in time period t; L_j are the Laguerre polynomials of order j; X_i is the value of i, where i can reflect coal ($_f$), CO_2 ($_c$), fuel oil ($_o$) or utilization rate ($_u$); F_t is the regression function estimated for each time period t; a_{ij} and c_n are coefficients estimated by the regression analysis.

[3] For financial options, Longstaff and Schwartz (2001) use for the regression exercise only results which are in-the-money. An option is in-the-money, if it would lead to a positive cash flow, when it is exercised immediately.or real options, it is not visible if the option is in-the-money with the information at a certain moment, because it depends on unknown future cash flows. First runs indicate that using only paths with a cash flow of zero or lower give better results and decrease the calculation time of the process, compared to a regression exercise using all simulation paths.

The value of the option to switch to another storage reservoir is determined in a similar way than the abandoning option. If the first reservoir is full, a switch to another storage reservoir can be made. If the option is not present or not exercised, the project has to be abandoned. It is valuable to exercise the option when the estimated future cash flows are higher than the costs for making a connection to the new storage reservoir. For this option, the regression exercise is based on all simulation results.

By comparing the project value without any options with the project value including one or multiple options, the option value is found. In addition, the probability and timing of abandoning, switching temporarily off the capture unit or connecting to another storage reservoir can be derived (Zhu and Fan, 2013).

5. Input data for the case study

In this article, all costs are expressed in €$_{2010}$ and corrected with the upstream capital cost index (IHS, 2014). However as this index is only valid for dollars, costs quoted in literature in Euros with another base year are first converted to dollars with the average exchange rate of the year where the costs are specific for. Subsequently, they are converted to $\$_{2010}$ with the relevant upstream capital cost index (IHS, 2014) and then back to Euros with the average exchange rate of 2010, which is 0.75 €$_{2010}$/$\$_{2010}$ (OANDA, 2014). Other economic assumptions are given in Table 2 while the correlation coefficients used for the different variables are given in Table 3.

The preference for ship or pipeline transport is analyzed for a case study under uncertainty. It is acknowledged that also the CAPEX, fixed OPEX and ABEX are uncertain. Nevertheless, in this study, it is assumed that they are known in order to assess clearly the influence of uncertainty in storage volume, utilization rate and commodity prices. The characteristics of the case study are given in Section 5.1 and the design and cost inputs for pipeline and ship transport are given in Sections 5.2 and 5.3, respectively.

5.1. Case study

The case study included in this article represents a typical coal power plant with a capture unit near a harbor. Although, the focus of this article is on the transportation chain, the capture unit as well as the storage facility are incorporated to be able to analyze when it is cost-effective to switch off the capture unit temporarily or abandon the CCS project. Hence, the capture as well as the storage part are included, but they are less detailed modelled than the transportation chain. The following assumptions for the case study were made:

- Three different design capacities are considered, representing a small demonstration, a large demonstration and a commercial coal-fired power plant with CCS. These design capacities are 133, 333 and 1330 tCO_2/h, which are equivalent to 1 MtCO_2/y, 2.5 MtCO_2/y and 10 MtCO_2/y with 7500 operating hours per year, respectively.
- It is assumed that there is a coal power plant with a CCS unit under construction. The specifications of this plant are based on a study of ZEP (ZEP, 2011a). By assuming a scale factor of 0.7 (Kuramochi et al., 2010), the investment costs for the CO_2 capture and compression unit are 215 M€, 408 M€ and 1076 M€ for design capacities of 1.0, 2.5 and 10 MtCO_2/y, respectively. The corresponding fixed OPEX are 5.6 M€, 11 M€ and 28 M€. Furthermore, the variable OPEX for the capture unit are 3 €/tCO_2 captured and the additional fuel use (including compression to 10–11 MPa) is 2 GJ_{pr}/tCO_2 captured. Due to the additional energy use induced by capturing CO_2, the amount of CO_2 avoided is lower than the

Table 3
Correlation coefficients between the different stochastic variables included in this study.

	CO_2 price	Coal price	Oil price	Electricity price	Utilization rate
CO_2 price	1	−0.46[a]	0.26[b]	0.39[b]	0[c]
Coal price	−0.46[a]	1	0.32[d]	0.56[a]	0[c]
Oil price	0.26[b]	0.32[d]	1	0.18[b]	0[c]
Electricity price	0.39[b]	0.56[a]	0.18[b]	1	0[c]
Utilization rate	0[c]	0[c]	0[c]	0[c]	1

[a] Based on Roques et al. (2008).
[b] Based on Abadie et al. (2014).
[c] In this study, the utilization rate of the coal power plant with CCS is strongly related to the penetration rate of renewables. Currently, renewables are subsidized and most renewables have a strong intermittent character. This implies that the utilization rate of the coal power plant is mostly determined by the electricity generated by the renewables, rather than by the prices for coal, CO_2 and electricity. Consequently, the correlation coefficients related to the utilization rate are set to zero.
[d] Based on Rothe (2011).

amount of CO_2 captured. For every tonne CO_2 captured, 0.81 t CO_2 is avoided[4].
- The costs for ramping up or down the capture plant have only a slight effect on the valuation and they are therefore not taken into account (Abadie, 2015; Fleten and Näsäkkälä, 2010).
- There is not much cost data available for the costs of decommissioning the CO_2 capture unit. It this study, it is assumed that the residual value equals the decommissioning costs (Malley and Zarider, 2012).
- It is assumed that after compression and drying, the CO_2 only contains traces of impurities and it behaves as pure CO_2.
- The CO_2 source is assumed to be located 10 km from the harbor, wherefrom the ships would depart or a suitable onshore–offshore connection could be made for the pipeline configurations. The outlet pressure of the onshore pipeline is set at 10 MPa and 20 °C (Yoo et al., 2013b). The onshore pipeline is designed by assuming a pressure drop of 30 Pa/m and a carbon steel grade of X80 (Knoope et al., 2014). The CAPEX for the onshore pipeline are 4.6, 7.1 M€ and 11 M€ for design capacities of 1.0 $MtCO_2$/y, 2.5 $MtCO_2$/y and 10 $MtCO_2$/y, respectively (Knoope et al., 2014). In this study, the ABEX for this onshore pipeline of limited length is assumed to be negligible. This assumption would not influence the choice between pipeline and ship, because both configurations include this pipeline.
- The distance from the harbor to the offshore reservoir is assumed to be 250 km, which is a typical distance to suitable large scale storage reservoirs on the North Sea, like from Kårstø to Utsira, from Rotterdam to J06A and from Peterhead to Brae (Apeland et al., 2011a; Austell et al., 2011; SCCS, 2009). In addition, the effect of a distance of 500 km is also investigated, which is approximately the distance from Rotterdam to (almost) depleted oil fields in Denmark (RCI, 2011). An additional reason to analyze 250 and 500 km is that the choice between pipeline and ship is not as straightforward as it would be for a distance below 100 km or above 1000 km.
- All CO_2 from the source is assumed to be transported to the same storage reservoir.
- Even after exploration of the storage reservoir, there will be uncertainty in the storage capacity, because perfectly mapping of the subsurface is neither possible nor desirable (Friedmann, 2007). Hence, part of the reservoir properties determining injectivity and capacity will be revealed during operation (Ringrose, 2013). The uncertainty in the storage capacity before actual site development is not known. It is stated that an accuracy of 30% should be realized in estimating the CO_2 storage capacity in geologic formations (NETL, 2013). By assuming that this reflects the 95% confidence interval of the normal distribution's probability density, the standard deviation is 15%. The average storage capacity is set at 100 Mt, which is considered to be the minimum capacity for economical offshore CO_2 storage (Wildgust, 2009).
- Due to geological factors, there are natural clusters of storage reservoirs, at least in the North Sea (SCCS, 2009; Van den Broek et al., 2010). Hence, the distance from one storage reservoir to another is expected to be less than the (assumed) 250 or 500 km from the harbor to the storage reservoir. The distance between storage reservoirs will be very case specific, but is assumed to be 25 km in this study.

5.2. Pipeline design and costs

The design and costs of the offshore pipeline are obtained from a cost minimization model developed in a previous study (Knoope et al., 2014). The outlet pressure of the offshore pipeline is set at 7 MPa, which is assumed to be high enough for direct injection (Apeland et al., 2011b). Increasing the pressure by installing pumping stations would be very expensive offshore, because a platform with an electricity connection is needed (IEA GHG, 2002). Consequently, if the inlet pressure of the offshore pipeline is higher than the outlet pressure of the onshore pipeline (10 MPa), a pumping station has to be installed before going offshore. For offshore pipelines and inlet pressures ranging from 10 to 35 MPa (in steps of 1 MPa), the most cost effective steel grade, required thickness and nominal pipe size is determined. The required thickness is based on maximum allowable operation pressure (MAOP), which is assumed to be 20% (instead of 10%) higher than the inlet pressure, to have some operational freedom to extend the length of the pipeline if the first storage reservoir is full[5]. For each inlet pressure, the total and levelized costs are calculated and the one resulting in the lowest levelized costs is selected. In this way, the optimal combination of inlet pressure, diameter and steel grade is found. The main outputs of the cost minimization model and costs data for the considered capacities and distances are given in Table 4.

If a connection has to be made to the next storage reservoir, the old platform would need to be decommissioned and a (new) platform needs to be installed at the new CO_2 injection location. In addition, a new pipeline section has to be added to the original pipeline to the new storage location. It is assumed that the CO_2 also has to be delivered to the new storage reservoir with a pressure of 7 MPa. To compensate for the pressure drop in the extended pipeline, the inlet pressure of the offshore pipeline has to be higher and additional pump capacity needs to be installed. This higher inlet pressure is no problem for the existing pipeline, because the MAOP was initially assumed to be 20% higher than the inlet pressure.

[4] In addition, CO_2 emissions are emitted downstream due to burning fuel oil, generating electricity and CO_2 that boils off. For the shipping solution, this will be about 2.5% of the total amount of CO_2 transported for a distance of 200 km (IEA GHG, 2004). These CO_2 emissions are ignored in this study.

[5] The MAOP always have to be 10% higher than the inlet pressure for safety reasons (Knoope et al., 2014). With a MAOP of 20% higher than the inlet pressure, extension of the pipeline with 25 km is possible without exceeding the MAOP requirement.

Table 4
Details and capital costs for the offshore pipeline solutions for the considered design capacities and distances.

	1 $MtCO_2/y$		2.5 $MtCO_2/y$		10 $MtCO_2/y$		Comment/source
	250 km	500 km	250 km	500 km	250 km	500 km	
Steel grade	X80	X80	X80	X65	X65	X80	[a]
Outer diameter (m)	0.22	0.27	0.32	0.41	0.61	0.61	[a]
Pressure drop (Pa/m)	51	15	40	12	24	24	[a]
Inlet pressure (MPa)	20	15	17	13	13	19	[a]
CAPEX pumping station[b] (M€)	2.8	1.8	3.8	2.3	4.8	14	Knoope et al. (2014)
Length depended CAPEX for offshore pipeline[d] (M€/km)	0.31	0.38	0.47	0.60	1.0	1.2	Knoope et al. (2014)
Fixed CAPEX for offshore pipeline[c] (M€)	35	35	35	35	35	35	Austell et al. (2011)
CAPEX platform[e] (M€)	68	68	68	68	68	68	Van den Broek et al. (2010)

[a] Outcomes of the cost minimization model of Knoope et al. (2014).
[b] Pumping costs are calculated with $I_{pump} = 74.3 \times W_{pump}^{0.58}$, where I_{pump} are the investment costs of pumping stations (k€) and W_{pump} is the capacity of pumping station per unit (kW_e).
[c] The fixed costs are for mobilization of the required pipelay barges, construction of the onshore–offshore connection and the connection of the pipeline to the injection facility. These costs are estimated at 35 M€ and independent of pipeline length (Austell et al., 2011).
[d] The length depended CAPEX for the offshore pipeline consist of material, labor and miscellaneous costs. For the cost formulas, we refer to previous work (Knoope et al., 2014).
[e] No platform is assumed to be present and, therefore, all surface facilities have to be constructed. The CAPEX for this are based on offshore aquifers and are estimated at 68 M€ (Van den Broek et al., 2010). The costs for site development (0–27 M€) and monitoring (0–1.7 M€) are excluded.

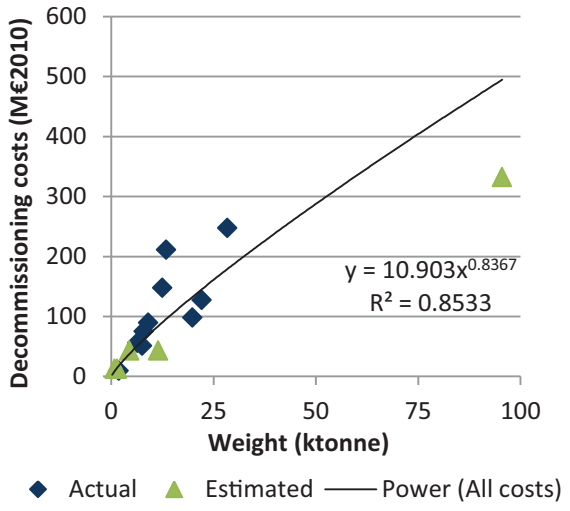


Fig. 3. Decommissioning costs for platforms. The following sources are used for the platform costs (Blacklaws et al., 2014; Davies et al., 2014; Energy Resource Technology (UK) Limited, 2012; Harvey and Walton, 2013; Maureen Owners, 2001; Shell, 2004; Sparreboom and Zant, 2014; TOTAL, 2011; Tucker et al., 2010; Tullow Oil, 2014).

The abandonment expenditures (ABEX) are based on reports submitted to the British Department of Energy and Climate Change (DECC), which regulate the decommissioning of offshore oil and gas installations and pipelines in the continental shelf of the United Kingdom (DECC, 2015). Redundant platforms at the North Sea have to be completely removed (DECC, 2011)[6]. In Fig. 3, an overview is given of the different cost estimations and realized costs for decommissioning of different platform weights. There is a relation between the decommissioning costs and the weight of the platform. In this study, the CO_2 injection is assumed to be conducted from a small platform, which weighs about 1.2 kt[7]. For this platform, the ABEX are estimated at 13 M€ with the derived power relation[8].

The pipelines are assumed to be decommissioned by flushing the pipelines with seawater and burying the ends of the pipelines (i.e., left in-situ). From the information available from the DECC, it is not possible to predict the decommissioning costs of pipelines by looking to the length, diameter or number of pipelines decommissioned. Hence, the costs relation derived for decommissioning offshore structures in the Gulf of Mexico was used, see Eq. (14) (Kaiser and Liu, 2014). The decommissioning costs depend on the water depth. In this study, a water depth of 100 m is assumed, which is comparable to the average depth of the North Sea (Ecomare, 2014).

For the pumping station, it is conservatively assumed that the decommissioning costs equal the residual value.

$$C_{decom_pipe} = 34{,}160 + 4043 \times WD + 1{,}355{,}429 \times OD + 7{,}919 \times L \quad (14)$$

where C_{decom_pipe} are the decommissioning costs of the pipeline (€$_{2010}$); WD is the water depth (m); OD is the outer diameter of the nominal pipe size (m) and L is the length of the pipeline (km).

5.3. Ship design and costs

In this study, CO_2 ship transport is based on near triple point conditions, meaning that the CO_2 is transported with a pressure of about 7 bar and about −50 °C. Different cost estimations for ships are available in literature (Apeland et al., 2011b; IEA GHG, 2004; Nam et al., 2013; Roussanaly et al., 2013ab; Skagestad et al., 2014). In Fig. 4, an overview is given and a power function is drawn based on all cost estimations[9]. In this study, the shipping costs of Roussanaly et al., (2013b) are used, because the costs are relatively close to the power function and, moreover, the fuel oil consumption of these ships are clearly stated. The fuel oil consumptions are 6.2, 5.7 and 5.4 $g_{fuel}/tCO_2/km$ for ship capacities of 25, 35 and 45 kt, respectively (Roussanaly et al., 2014). Fixed OPEX and CAPEX for the ships as well as for the other components are given in Table 5. The ships and floating vessels are assumed to be only available in

[6] Exceptions can be made for platforms with a concrete structure or large scale steel platforms with a weight of more than 10 kt, if they are installed before 1999 (DECC, 2011).
[7] This is about the average weight of the five smallest platforms (Juliet-P, Mike, and November platforms of the Indefatigable field; Welland, and Hron & Wren platform). These platforms support on average four wells.
[8] By conducting a sensitivity analysis by removing one data point at the time, the ABEX would vary between 11 and 15 M€. This range is within the 30% uncertainty range often assumed present in feasibility studies (McCoy, 2012; U.S. Environmental Protection Agency & U.S. Army Corps of Engineers, 2000).
[9] Nam et al. (2013) give shipping costs for different target pressures and speeds. In Fig. 4, the costs are given for a target pressure of 5 bars and a speed of 15 knots. The capacities of the ships were given in cubic meters for Nam et al. (2013), Apeland et al. (2011b), and Skagestad et al. (2014). These are converted with a density of 1155 kg/m^3.

Table 5
Selected capital costs of the different components for ship transport.

Component	Base scale	Scale unit	Scale factor	Base costs (M€$_{2010}$)	Max. size of unit	Fixed O&M cost	Source
Liquefaction[a]	20	$MtCO_2/y$	0.90	147	n.a.	5%	ZEP (2010)
Temporary storage at the harbour[b]	1000	m^3	1.0	1.0	n.a.	5%	ZEP (2010)
Loading equipment[c]	3	$MtCO_2/y$	0.0	9.5	5	2%	Apeland et al. (2011b)
Ships[d]	25 (21,825)	$ktCO_2$ (m^3)	n.a.	44	n.a.	2.2 M€/y	Roussanaly et al. (2013a,b)
	35 (30,555)	$ktCO_2$ (m^3)	n.a.	52	n.a.	2.5 M€/y	
	45 (39,285)	$ktCO_2$ (m^3)	n.a.	60	n.a.	2.7 M€/y	
Floating vessel[e]	25	$ktCO_2$	n.a.	23	n.a.	5%	Business News (2014), Simic (2013)
	35	$ktCO_2$	n.a.	28	n.a.		
	45	$ktCO_2$	n.a.	33	n.a.		
Offloading system[f]	1200	tCO_2/h	0.29	30	1200	5%	Apeland et al. (2011b), Skagestad et al. (2014)
Conditioning[g]	350	tCO_2/h	0.56	2.0	n.a.	5%	Apeland et al. (2011b)

[a] The IEA GHG (2004) estimated the costs for a stand-alone liquefaction unit receiving 6.2 Mt pre-pressurized CO_2 per year on 28 M€. ZEP (2010) estimated the capital costs including construction interest for a 2.5 $MtCO_2/y$ and a 20 $MtCO_2/y$ liquefaction unit on 22.7 and 147 M€, respectively. These costs are used to estimate the scale factor of 0.9. By using this derived sale factor, the costs of ZEP for a 6.2 $MtCO_2/y$ unit would be 51 M€. Hence, the costs of ZEP are 80% higher than from IEA GHG. The costs of ZEP are used in this study because they are more recent and it is clearly stated that also the recompression of the gaseous CO_2 is included. The fixed O&M costs are not stated in the study of ZEP and are therefore based on IEA GHG (2004).

[b] The cost of temporary storage in the harbor is based on utilizing a floating storage vessel permanently located between the ship and the quay (ZEP, 2010). This option is expected be the least cost alternative when sufficient harbor space is available, because construction can take place outside the harbor (ZEP, 2010). Therefore, this storage option is also used in this study. The volume of temporary storage should be similar as the cargo volume of the largest ship in use (Skagestad et al., 2014; ZEP, 2010). This assumption is also used in this study.

[c] Not many cost estimations are available for loading equipment. It is assumed that loading equipment is already installed in the harbor (ZEP, 2010) or that the costs are negligible (Decarre et al., 2010). Apeland et al. (2011b) assume that the costs for loading equipment are 9.5 M€ for handling mass flows of 1, 3 or 5 $MtCO_2/y$. Skagestad et al. (2014) mention substantially lower costs of 1.1 M€ for loading 0.8 $MtCO_2/y$ and IEA GHG (2004) estimated the costs on 7.5 M€$_{2010}$ for loading 6.2 $MtCO_2/y$. In this study, the cost estimation of Apeland et al. (2011b) is conservatively used and the maximum unit size is assumed to be 5 $MtCO_2/y$.

[d] For ships, three sizes are assumed to be available and no intermediate sizes can be purchased.

[e] A floating vessel is used as offshore temporary storage and as location for the conditioning equipment. This can be a converted (old) ship (Choi and Chang, 2011). Similar to the onshore temporary storage, the size of the floating vessel is assumed to be similar as the size of the largest ship in use. Not much cost information is publically available on the costs of floating vessels. It is known that the conversion costs for a 40,000 m^3 oil tanker were 25 M€ (Business news, 2014) and 40 M€ for an oil tanker of almost 80,000 m^3 (Simic, 2013). Based on these data points, a scale factor of 0.69 can be derived. This scale factor and the two data points are used to estimate the conversion costs. In addition to the conversion costs, an old ship has to be purchased. The purchasing cost of an old ship of 40,000 m^3 is 10 M€ (Simic, 2013), which is approximately 15% of its initial value (see Fig. 4). Hence, it is estimated that old ships can be purchased of 15% of their initial value. The O&M costs are assumed to be the same percentage as for the temporary storage near the harbor and the offshore loading system.

[f] The floating vessel is connected to the storage reservoir with an internal dis-connectable submerged turret loading system (STL) and a spread mooring system. The CAPEX for a large STL system capable of handling 1200 tCO_2/h are estimated at 30 M€ and the fixed OPEX at 5% of the CAPEX by Apeland et al. (2011b). Skagestad et al. (2014) estimated the investment costs at 21.9 M€ for a STL system with a throughput of 400 tCO_2/h. With these two points the scale factor is estimated at 0.29, which indicates very large economies of scale. No other cost estimations are found for a STL in publically available sources and therefore this relation is used. To avoid unrealistic large economies of scale, the maximum scale is conservatively set at 1200 tCO_2/h.

[g] In this study, the minimal temperature and pressure at which the CO_2 can be injected in the reservoir is set at 0 °C and 7 MPa (Apeland et al., 2011b). To reach these conditions for ship transport, two heat exchangers and three pumps are installed on the floating vessel. The energy required for this is provided by burning fuel oil, which has to be supplied to the injection site. The CAPEX for the entire conditioning process are estimated at 2 M€ for a flow of 350 tCO_2/h and 4 M€ for a flow of 1200 tCO_2/h (Apeland et al., 2011b). This leads to a scaling factor of 0.56.

three sizes. The other components can be scaled to the required size with the following equation:

$$C_1 = C_0 \times \left(\frac{S_1}{S_0}\right)^y \quad (15)$$

where C are the costs of the component; S is the size of component; y is the scale factor; and the subscripts 1 and 0 refer to the required and reference scale, respectively.

After the ship has run his economic lifetime, the residual value of the ship is estimated to be 200 € per tonne, based on the scrap iron price (Return, 2011a,b). A ship transporting 16,000 m^3 CO_2 weights about 19.5 kt and a ship of 32,000 m^3 about 39 kt (Apeland et al., 2011a). Hence, the residual value would be 3.9 M€ and 7.8 M€, respectively. This is about 10% of the initial investment costs, which is similar to the residual value of 10% stated by Bingsong et al. (2014). In this study, it is assumed that the residual value of ships is at least 10%. However, if the ship is sold already after a few years of operation it is probably worth more, because it can be used for transport of other commodities. Aspelund et al. (2006) estimated the residual value of a ship on 35% after 15 years. Assuming linear depreciation, and extrapolating the trend of Aspelund et al., the ship is assumed to depreciate to 10% of the initial investment after 21 years. After 21 years, the residual value is assumed to remain 10%. For the other components (liquefaction unit, temporary storage unit, floating vessel, heaters and pumps), it is assumed that the decommissioning costs equal the residual value.

The optimal ship configuration is determined by a simple cost minimization model made in Excel. First, the minimum required number of ships is calculated for each ship size, by taking into account the (un)loading time, the availability and the sailing speed. Second, the required capacities, CAPEX and OPEX are calculated for each component. Subsequently, the configuration leading to the lowest total costs is selected. In this study, it is not taken into account that configurations of ships with different sizes can be combined.

If the storage reservoir is full, the ships can sail to another reservoir, which is further away from the harbor. The storage vessel and the offloading system has to be disconnected from the old storage location, towed to and moored at the new storage reservoir. Unfortunately, not much cost data for this operation is available. The realized costs for disconnecting and towing a floating production, storage and offloading (FPSO) unit to another location are estimated to be 9.1 M€ (Premier Oil, 2012). To incorporate also the reinstallation of the floating vessel, the costs are assumed to be 10 M€ in this study.

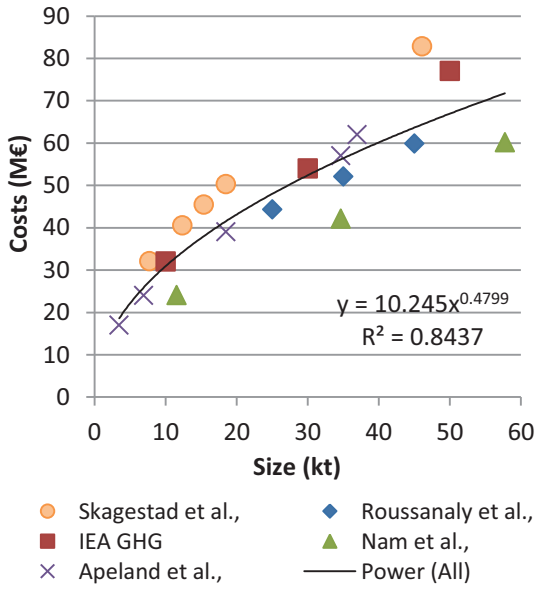


Fig. 4. CAPEX for CO_2 ships available in literature (Apeland et al., 2011b; IEA GHG, 2004; Nam et al., 2013; Roussanaly et al., 2013ab; Skagestad et al., 2014).

6. Results

The results for the net present value (NPV) approach, including a sensitivity analysis, are presented for the ship and pipeline solution in Section 6.1. In Section 6.2, the results of the real option approach are given and compared with the NPV approach.

6.1. Pipeline versus ships with the NPV approach

In Fig. 5, the levelized transport and storage costs are given for different ship sizes and compared to the costs of pipeline transport for a design capacity of 1, 2.5 and 10 $MtCO_2$/y over 250 km. The costs directly related to the ship represent about 20–30% of the total levelized shipping costs. The other 70–80% of the costs is mainly related to the harbor facilities, liquefaction unit, offloading and conditioning equipment. The liquefaction costs are relatively independent of mass flow, due to the constant energy consumption per tonne of CO_2, while the costs of the harbor facilities, ships and offloading and conditioning equipment decrease with increasing mass flows. Overall, the levelized costs for shipping increase with ship size for all three mass flows with a distance of 250 km. The reason for this is that the ships can transport with a distance of 250 km, maximal 5.2, 7.5 and 9.4 $MtCO_2$/y for ship sizes of 25, 35 and 45 kt, respectively. Hence, one ship is needed for 1 and 2.5 $MtCO_2$/y and two ships are needed for 10 $MtCO_2$/y, independent of the size of the ships. If the distance increases to 500 km, the 1 and 2.5 $MtCO_2$/y configurations remain based on one ship of 25 kt, while for the 10 $MtCO_2$/y configuration it is most cost-effective to be based on two ships of 35 kt.

For 250 km, ships realize lower levelized transport and storage costs than pipelines for a design capacity of 1 Mt/y, while pipelines realize lower levelized costs for a capacity of 10 Mt/y, see Fig. 5. For a capacity of 2.5 Mt/y, pipelines also realize lower levelized costs, but the difference with ships is only 2%.

In Tables 6 and 7, the economic figures are given for the most cost-effective pipeline and ship configuration for 1, 2.5 and 10 $MtCO_2$/y over 250 km and 500 km, respectively. Two different sets of economic figures are given for a design capacity of 10 $MtCO_2$/y, one incorporating the fact that the CO_2 storage reservoir has a limited capacity of 100 Mt and the other representing a project with a fixed duration of 25 years. The consequences of having only a limited storage capacity available are significant. It increases the overall levelized costs with 16% and 13% for the pipeline and shipping configuration, respectively. In addition, it changes the preferred transportation mode from pipeline to ship transport in the 500 km case. Moreover, it changes the NPV from positive to negative for the pipeline configuration with a distance of 250 km. For the rest of the article, only the cases with limited storage capacity are considered.

Two different trends can be noticed from Tables 6 and 7. First, the levelized costs decrease with increasing mass flows for ship as well as for pipeline transport, but the economies of scale are larger for pipeline than for ship transport. Second, the levelized costs increase with distance, but the shipping costs are less sensitive to distance than the pipeline costs. Consequently, ship transport is more cost-effective than pipeline transport for a distance of 250 km for the smallest design capacity and for all capacities on a distance of 500 km (at least with limited storage capacity). However, none of the considered cases has a positive NPV with a limited storage capacity of 100 Mt and an initial CO_2 price of 35 €/t, which increase with 3% per year.

A sensitivity analysis was performed for the NPV of the entire CCS chain. In Fig. 6, the results of this analysis are presented for 2.5 $MtCO_2$/y and 10 $MtCO_2$/y over 250 km, but similar relations are seen for the smallest capacity and a distance of 500 km. The NPVs of both the shipping and pipeline configuration are sensitive to changes in the initial CO_2 price, utilization and interest rate. The NPV turns positive in the 10 $MtCO_2$/y pipeline and shipping cases with a distance of 250 km, when the initial CO_2 price increases to 39 and 43 €/t, respectively. In addition, the project value of the 10 $MtCO_2$/y—250 km pipeline case turns positive if the interest rate decreases to 6.6%. The decision between ship and pipeline transport is the most sensitive for the 2.5 $MtCO_2$/y—250 km case. For this case, the preference switches from pipeline to ship transport if the electricity price reduces to 50 €/MW h, the utilization rate decreases to 78% or the interest rate increases to 11%. Also, the 10 $MtCO_2$/y—500 km case is sensitive to changes. For this case, the preference switches to pipeline transport if the electricity price increase from 60 to 97 €/MW h.

6.2. Pipeline versus ships with real option approach

In this study, three options were considered, namely the option to switch to another reservoir, to abandon the project, and the option to switch temporarily off the capture unit. In Tables 8 and 9, the results of the ROA with and without options are given for 250 and 500 km, respectively. Without options, similar trends can be seen as by the NPV approach, namely the costs of ship transport are less sensitive to length and mass flow than pipeline transport. In addition, for the 1.0 and 2.5 $MtCO_2$/y cases the preferred transportation mode does not change as well as the fact that investing with the assumed initial price of 35 €/tCO_2 is not profitable. For the 10 $MtCO_2$/y cases, things change. With a distance of 250 km, the 10 $MtCO_2$/y pipeline case has a positive project value with the ROA, meaning that investing is profitable with an initial price of 35 €/tCO_2. For 500 km, the preferred transportation mode switches from ship to pipeline, but the project value remains negative. These changes are the consequence of using the risk free rate of 5% instead of the discount rate of 10%.

In the following sections, the effects of the options are first analyzed separately and then the effect of the three combined options is evaluated.

6.2.1. Option to switch to another storage reservoir

The switching option is not very relevant for the 1.0 and 2.5 $MtCO_2$/y cases, because the CO_2 can be stored in the first storage reservoir for 25 years in 100% and >99.9% of the cases,

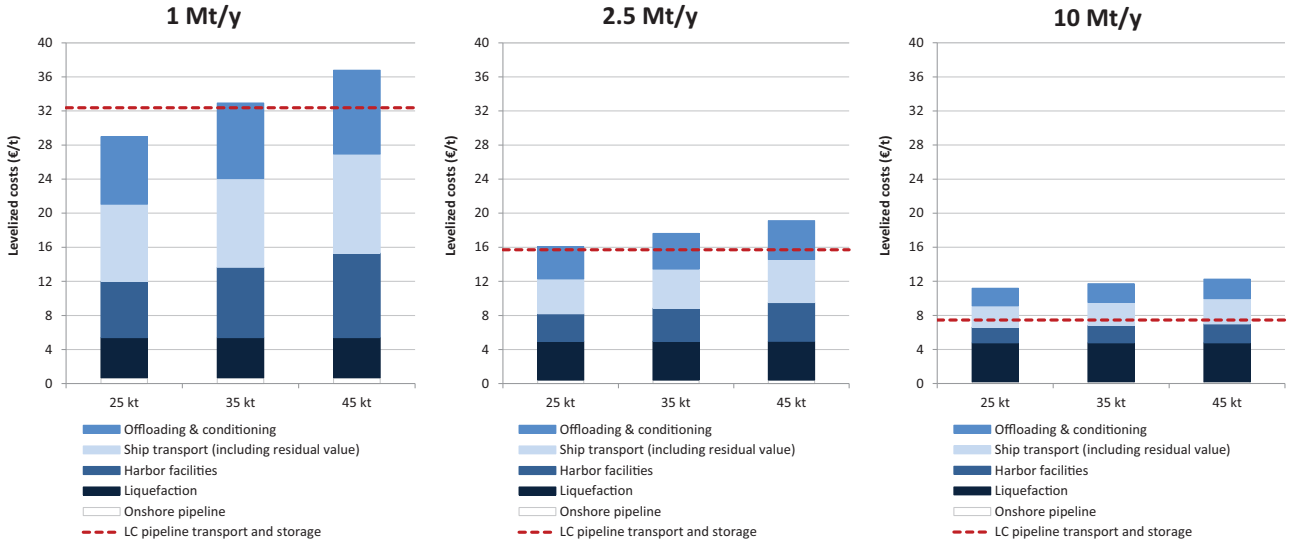


Fig. 5. Estimated levelized transport and storage costs calculated with the NVP approach of pipeline (dotted line) and ship transport (bars) for a design capacity of 1.0 $MtCO_2$/y (left); 2.5 $MtCO_2$/y (middle) and 10 $MtCO_2$/y (right) over 250 km.

Table 6
Results of the NPV approach for the pipeline and ship configuration for a distance of 250 km.

	1.0 $MtCO_2$/y		2.5 $MtCO_2$/y		10 $MtCO_2$/y—limited storage capacity		10 $MtCO_2$/y—fixed project duration of 25 y	
	Pipeline	Ship	Pipeline	Ship	Pipeline	Ship	Pipeline	Ship
NPV whole CCS project (M€)	−301	*−275*	*−273*	−280	*−233*	−456	*333*	−1.3
Levelized transport & storage costs (€/tCO_2)	32.4	*29.0*	*15.7*	16.1	*7.5*	11.2	*6.1*	10.4
Overall levelized costs (€/tCO_2)	75.0	*71.6*	*50.0*	50.4	*36.5*	40.3	*31.4*	35.8
Required initial CO_2 price (€/tCO_2)	73.4	*70.1*	*48.9*	49.3	*39.2*	43.2	*30.7*	35.0

Table 7
Results of the NPV approach for the pipeline and ship configuration for a distance of 500 km.

	1.0 $MtCO_2$/y		2.5 $MtCO_2$/y		10 $MtCO_2$/y—limited storage capacity		10 $MtCO_2$/y—fixed project duration of 25 y	
	Pipeline	Ship	Pipeline	Ship	Pipeline	Ship	Pipeline	Ship
NPV whole CCS project (M€)	−425	*−281*	−478	*−294*	−609	*−531*	*−57.2*	−92.9
Levelized transport & storage costs (€/tCO_2)	48.6	*29.8*	26.4	*16.8*	13.7	*12.4*	*11.2*	11.6
Overall levelized costs (€/tCO_2 stored)	91.2	*72.4*	60.7	*51.1*	42.8	*41.5*	*36.5*	37.0
Required initial CO_2 price (€/tCO_2)	89.3	*70.9*	59.4	*50.0*	45.9	*44.5*	*35.7*	36.6

respectively. Consequently, the value of switching is zero for a design capacity of 1.0 $MtCO_2$/y and close to zero for a design capacity of 2.5 $MtCO_2$/y for both transportation modes, see Tables 8 and 9. In the 2.5 $MtCO_2$/y case, the option value is about twice as high for the ship than for the pipeline configuration, due to the lower investment costs required for the ship configuration combined with the limited time available to earn back the investment.

For the 10 $MtCO_2$/y case, the option to switch to another reservoir is very relevant, because the CO_2 can only be stored in the first storage reservoir for on average 11 years. Note that even with the switching option, the CCS project has to be abandoned with the 10 $MtCO_2$/y configuration in almost 40% of the runs before year 25, because both storage reservoirs are full. In Fig. 7, the cumulative probability that a switch to another storage reservoir has

Table 8
The project value (in M€) with and without options for the pipeline and ship solution for a distance of 250 km[a].

	1.0 $MtCO_2$/y		2.5 $MtCO_2$/y		10 $MtCO_2$/y	
	Pipeline	Ship	Pipeline	Ship	Pipeline	Ship
Project value without options	−223	*−208*	*−27.5*	−47.7	*130*	−142
Switch to another reservoir	−223 (0)	*−208 (0)*	*−27.5 (0.003)*	−47.7 (0.006)	*1307 (1177)*	919 (1061)
Switch off the capture unit	−223 (0.5)	*−206 (2.2)*	*−26.4 (1.1)*	−42.3 (5.4)	*130 (0)*	−142 (0)
Abandon the project	−213 (10.6)	*−177 (31.0)*	*−17.0 (10.6)*	−20.6[b] (27.1)	*134 (3.6)*	−120 (22.4)
All options combined	−213 (10.6)	*−177 (31.0)*	*−16.9 (10.6)*	−20.7[b] (27.0)	*1313 (1183)*	948 (1090)

[a] The option values of the different options are given in brackets in M€. The project and option value of the transportation mode resulting in the highest project value is presented in italics.

[b] In the 2.5 Mt pipeline configuration, the option value for the only abandoning option is 6 k€ higher than the option value for all options combined. This is caused by the slightly higher share (0.01%) of incorrect abandoning decision in the latter.

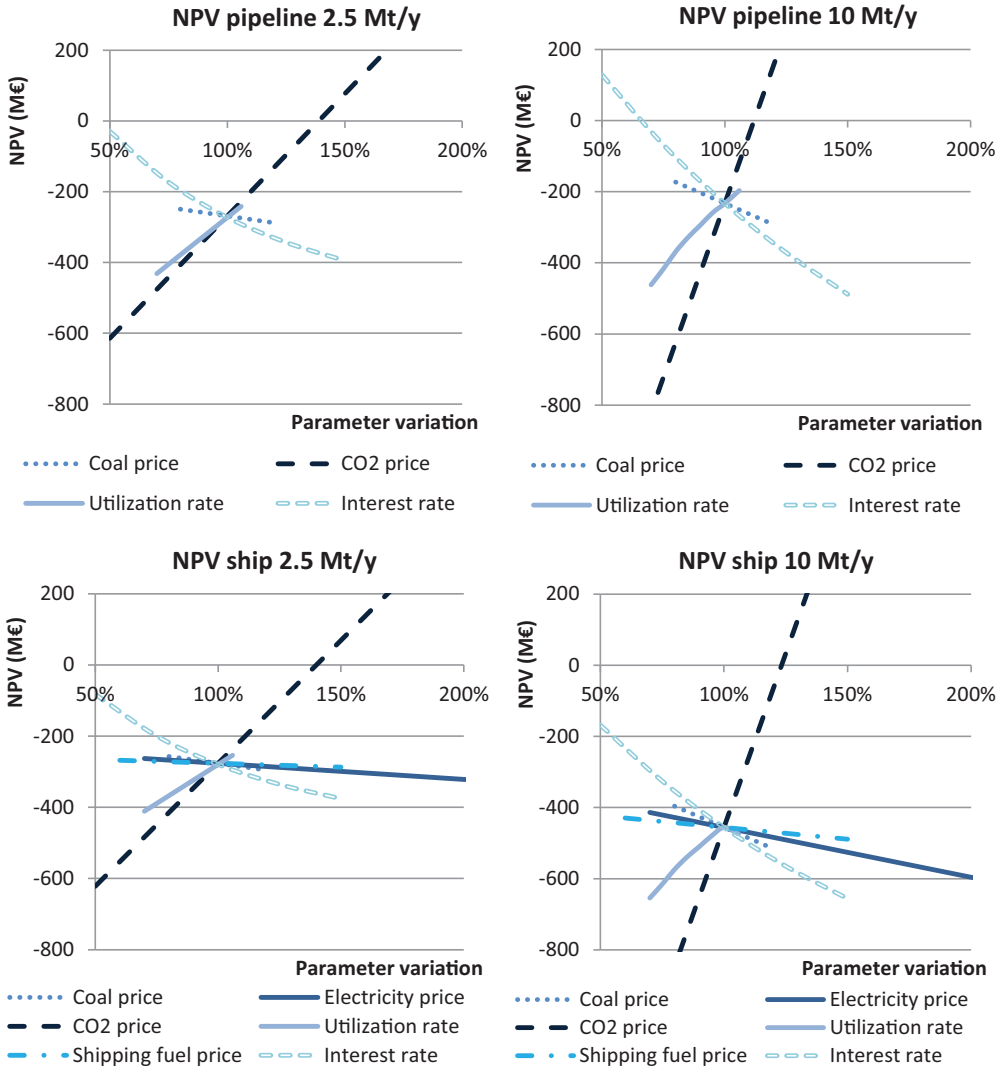


Fig. 6. Sensitivity analysis of the NPV for the entire CCS chain based on pipeline (upper) or ship transport (lower) for a design capacity of 2.5 $MtCO_2$/y (left) and 10 $MtCO_2$/y (right) over 250 km.

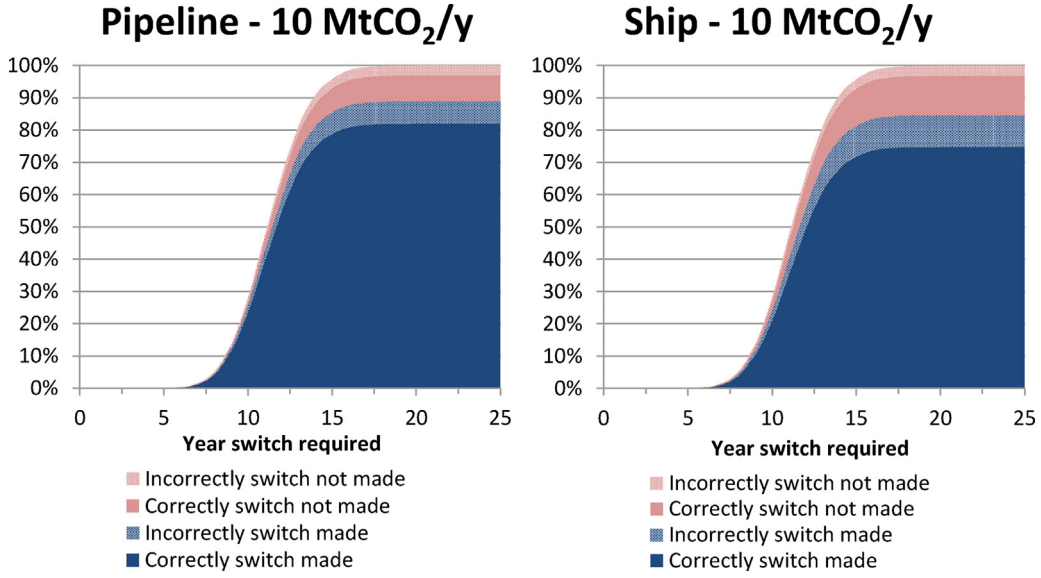


Fig. 7. Cumulative probability that a switch to another storage reservoir has to be made for the 10 $MtCO_2$/y pipeline (left) and ship (right) configuration over 250 km. A division is made between the probability that a switch to another reservoir is made (dark blue), and the probability that the switch to another reservoir is not made and that the CCS project is abandoned (light red). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

to be made is given for the 10 $MtCO_2$/y pipeline and ship configuration over a distance of 250 km. A division is made between a correct and incorrect switching decision. The project is stated to be correctly switched, if the project value is higher than it would be without making the connection to another storage reservoir. Similarly, the project is stated to be incorrectly switched, if the project value would be higher if a switch to another reservoir was not made. It can be assessed that the option to switch to another storage reservoir is more frequently made for the 10 $MtCO_2$/y pipeline than for the ship configuration, namely a probability of 89% against 85% for a distance of 250 km. The probability that the switch is incorrectly made is about 7%-points and 10%-points, respectively.

The reason that the option to switch to another reservoir is more frequently exercised for the pipeline than for the ship configuration is the lower OPEX of the former. This lower OPEX results in higher expected revenues after the switch is made. The higher expected revenues more than compensate the higher switching costs for the pipeline (129 M€$_{2010}$) compared to the ship configuration (10 M€$_{2010}$). Consequently, the value for the option to switch to another reservoir is about 10% higher for the pipeline than for the ship configuration with a capacity of 10 $MtCO_2$/y, as is shown in Tables 8 and 9. By incorporating the value of the switching option, all 10 $MtCO_2$/y cases have a positive project value. However, the project value of the pipeline remains higher for both distances.

If the distance increase from 250 km to 500 km, the option to switch to another storage reservoir becomes less valuable. The expected revenues after the switch are lower and, consequently, the connection to another storage reservoir is made less often. For instance, the cumulative probability is reducing from 89% to 87% in the 10 $MtCO_2$/y pipeline configuration and from 85% to 82% in the 10 $MtCO_2$/y shipping configuration, if the distance increase from 250 to 500 km.

6.2.2. Option to switch off the CO_2 capture unit temporarily

The costs associated with switching the capture unit on and off are assumed to be negligible. Hence, when the CO_2 price does not compensate the variable capture, transport and storage costs the CO_2 capture unit is switched off. The probability that the variable CO_2 capture, transport and storage costs are below the CO_2 price is about 3% and 7–10% of the runs for the pipeline and ship configuration, respectively. The option to switch off the CO_2 capture unit is more often exercised with the ship configuration, because the variable costs are about 50% higher for the ship than for the pipeline configuration. So, the option to switch off the CO_2 capture unit is more valuable for the ship than for the pipeline solution. In addition, the option value increases with distance for the shipping configurations because the variable OPEX also increases with distance. For the pipeline configurations almost no changes arise in the variable OPEX and consequently, the option value of switching off the capture unit for the pipeline configurations is unaffected by distance.

For 1.0 and 2.5 $MtCO_2$/y cases, the value of the option is about five times as high for the ship than for the pipeline configurations, see Tables 8 and 9. However, this difference is not enough to make ship transport more cost-effective than pipeline transport for the 2.5 $MtCO_2$/y case over 250 km. Furthermore, the option value is in none of the analyzed cases high enough to turn the project value positive.

For 10 $MtCO_2$/y, it is not cost-effective to switch off the CO_2 capture unit, if the option is evaluated separately and the entire lifetime of the project is taken into account. The reason for this is that if the CO_2 capture unit is switched off, no CO_2 is stored into the storage reservoir. This implies that the fixed costs have to be paid longer. These additional fixed costs are on average higher than the cost savings made by switching temporarily off the capture unit, especially when CO_2 prices drop below the variable capture, transport and storage costs for a longer period of time. Overall, the option to temporarily switch off the CO_2 capture unit does not have an additional value for the 10 $MtCO_2$/y cases, at least when it is analyzed without the option to abandon the project before the reservoir is full.

6.2.3. Option to abandon the project

In Fig. 8, the cumulative probabilities are given that a project is abandoned before the end of the lifetime for a distance of 250 km. A distinction is made between paths, which have to be abandoned because the reservoir is full and projects that are correctly or incorrectly abandoned before the reservoir is full. The project is stated to be correctly abandoned if the total project value after abandoning is higher than it would be if the project was continued. The ratio between correctly and incorrectly abandoned projects is about 85:15 for the pipeline and ship configurations.

The cumulative probability that CCS projects are abandoned before the reservoir is full is lower for pipeline than for ship transport, see Fig. 8. Abandoning the ships is more attractive because of the higher OPEX combined with the residual value of the ships. Furthermore, the abandoning probability is increasing with decreasing mass flows and with longer distances. This is mainly caused by the overall higher average project value of configurations handling larger mass flows over shorter distances. Overall, the probability that the CCS project is abandoned before the reservoir is full varies between 11% for the 10 $MtCO_2$/y—250 km pipeline configuration till 64% for the 1.0 $MtCO_2$/y—500 km ship configuration.

In Tables 8 and 9, the values of the abandoning option are given. The option to abandon the project is more valuable for the ship than for the pipeline configurations and more valuable for larger distances. The large difference between the option value for the 10 Mt/y ship configurations over 250 and 500 km is caused by the lower residual value of the 25 kt ships, which are used in the 250 km configuration, compared to the 35 kt ships, which are used in the 500 km configuration. For all analyzed cases, the option to abandon does neither turn negative project values into positive ones nor does it change the preferred transport mode.

Table 9
The project value (in M€) with and without options for the pipeline and ship solution for a distance 500 km[a].

	1.0 $MtCO_2$/y		2.5 $MtCO_2$/y		10 $MtCO_2$/y	
	Pipeline	Ship	Pipeline	Ship	Pipeline	Ship
Project value without options	−356	*−217*	−246	*−70.0*	*−258*	−278
Switch to another reservoir	−356 (0)	*−217* (*0*)	−246 (0.002)	*−70.0* (*0.005*)	*888* (*1146*)	734 (1012)
Switch off the capture unit	−356 (0.5)	*−215* (*2.8*)	−245 (1.1)	*−63.3* (*6.7*)	*−258* (*0*)	−278 (0)
Abandon the project	−342 (13.8)	*−184* (*33.5*)	−232 (13.8)	−39.9 (30.1)	*−253* (*4.9*)	−246 (32.6)
All options combined	−342 (13.8)	*−183* (*33.6*)	−232 (13.8)	*−39.6* (*30.3*)	*896* (*1154*)	771 (1050)

[a] The option values of the different options are given in brackets in M€. The project and option value of the transportation mode resulting in the highest project value is presented in italics.

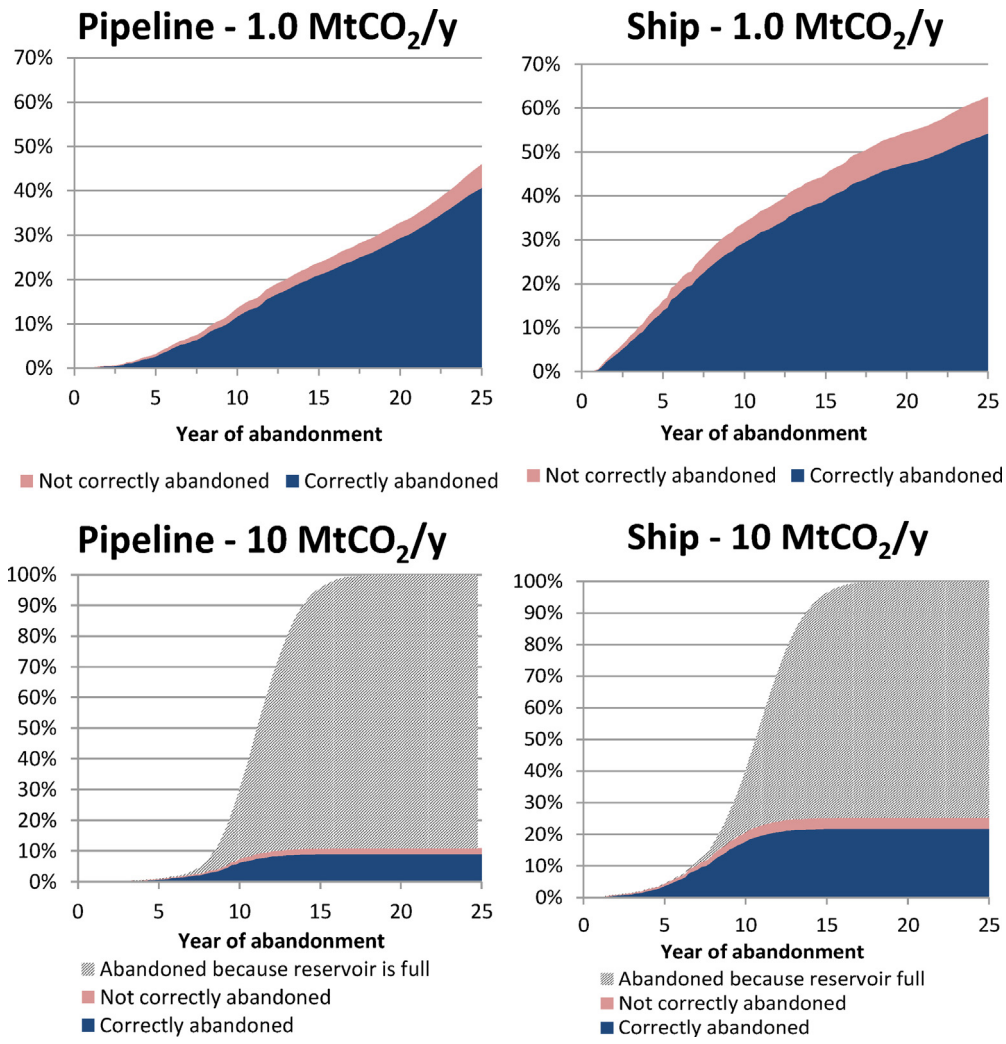


Fig. 8. Cumulative probability that a CCS project is abandoned before the end of lifetime and before the reservoir is full, when only the abandoning option is present for the pipeline (left) and ship (right) configuration with a design capacity of 1.0 $MtCO_2$/y (upper) and 10 $MtCO_2$/y (below) over 250 km.

6.2.4. All three options combined

In Tables 8 and 9, the values of all options separately and combined are given. For the 1.0 and 2.5 $MtCO_2$/y cases, the option value of all three options combined is less than the sum of the individual options. The main reason for this is that instead of switching off the capture unit for a long time, the project is now abandoned. Consequently, the probability that the option is exercised to temporarily switching off the capture unit is decreasing, for instance, from 7.8% to 0.15% and from 2.9% to 0.02% in the 2.5 $MtCO_2$/y ship and pipeline configuration, respectively, with a distance of 250 km. In addition, the probability that the CCS project is abandoned is slightly decreasing with 0.1%-points and 0.2%-points, respectively.

In the 10 $MtCO_2$/y cases, the value of all options combined is higher than the sum of the individual options, see Tables 8 and 9. The reason for this is that now it is profitable in several cases to switch temporarily off the capture unit, because this does not automatically mean that the lifetime of the project increases. Moreover, the project can also be abandoned after a switch is made to the second storage reservoir. In Fig. 9, the abandonment probabilities are given for the pipeline and ship configuration for a design capacity of 10 $MtCO_2$/y and 250 km. By comparing the pipeline and ship configuration, it can be assessed that the ship configuration has a higher abandonment probability before the storage reservoir is full, while the pipeline configuration is more frequently abandoned when the first reservoir is full and a switch has to be made to another reservoir. Note that still the switch to another reservoir is more frequently made with the pipeline (90%) than with the ship configuration (84%). Overall, the cumulative probability that the CCS project is abandoned before year 25 decreases due to the option to switch to another field, as can be assessed by comparing Fig. 8 with Fig. 9. For instance, in the 10 $MtCO_2$/y—250 km pipeline case, the probability of abandoning decrease from 100% to 56%, if also the options are present to switch to another reservoir and switch off the capture unit.

The option values for all three options combined are very significant for all configurations. Especially, the 10 $MtCO_2$/y cases have very high option values ranging from 1050 to 1183 M€. These values are mainly driven by the option to switch to another storage reservoir. As a consequence, the project value is positive for all 10 $MtCO_2$/y configurations, when all three options are taken into account. However, the preferred transportation mode does not change and pipeline transport remains the best option in all 10 $MtCO_2$/y cases. For the smaller design capacities, the option values are also significant with amounts of 10–14 M€ for the pipeline and 27–34 M€ for the shipping configurations. These option values are mainly driven by the abandoning option. For design capacities of 1.0 and 2.5 $MtCO_2$/y, the option value of all options combined neither changes the preferred transportation mode nor turns the project value positive.

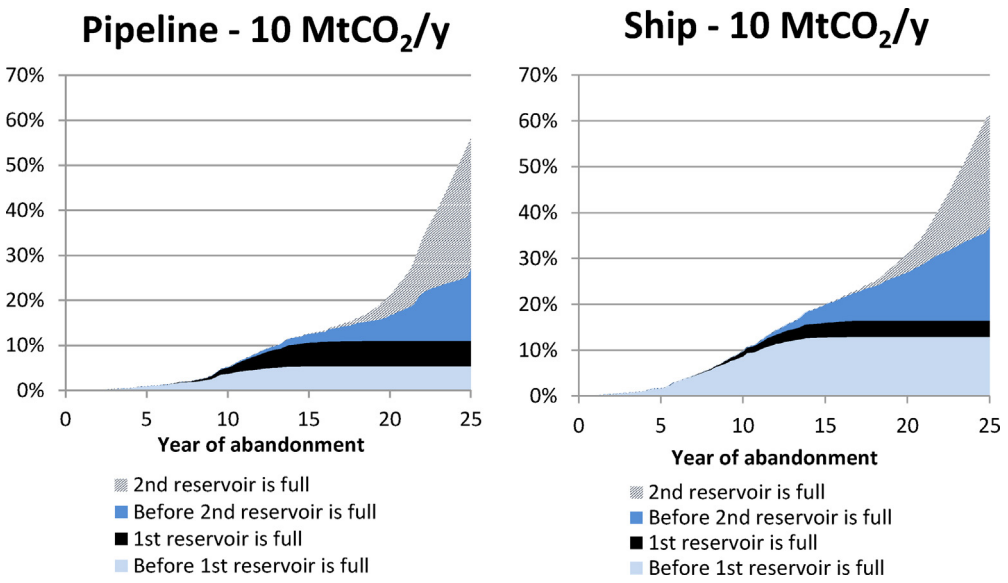


Fig. 9. Cumulative probabilities that a CCS project is abandoned for the pipeline (left) and ship (right) configuration with a design capacity of 10 $MtCO_2$/y over 250 km, when the options to temporarily switch off the capture unit, to abandon the project and to switch to another storage reservoir are present. A division is made between projects, which are abandoned because the first or second reservoir is full or before or after a switch is made to another storage field.

7. Discussion

7.1. Comparison with the literature

In this study, the shipping costs are based on several assumptions. To assess whether the resulting costs are similar to the ones stated in literature, the calculated shipping costs are compared with the cost estimations of Roussanaly et al. (2013a) and European Technology Platform for Zero Emission Fossil Fuel Power plants (ZEP, 2010). A comparison of the used pipeline cost model with other cost models available in literature is presented elsewhere (Knoope et al., 2014).

Roussanaly et al. (2013a) estimated the costs for ship transport over 480 km without liquefaction on 7.7 €/tCO_2 for transporting 15.45 $MtCO_2$/y to an onshore harbor. The cost model in this study predicts shipping costs (without liquefaction and submerged turret loading system) at 5.4 €/tCO_2 for a similar mass flow and distance[10]. There are three main reasons for this difference of about 40%. First, the fuel oil price in the study of Roussanaly et al. (2013a) is 790 €/t_{fuel} instead of 500 €/t_{fuel}. Implementing this higher fuel oil price increases the levelized costs to 6.4 €/tCO_2 and decreases the difference to about 20%. Second, the costs of Roussanaly et al. (2013a) include a higher total harbor fee of 2 €/t because two harbors are accessed. Thirdly, after delivering the CO_2, the pressure of the CO_2 is increased to 20 MPa instead of 7 MPa. Correcting for this outlet pressure and higher harbor fee, result in estimated levelized costs of 8.5 €/tCO_2, which is only slightly higher than the cost estimation of Roussanaly et al. (2013a)[11].

The European Technology Platform for Zero Emission Fossil Fuel Power plants (ZEP) estimated the levelized cost at 14 and 15 €/tCO_2 for transporting 2.5 $MtCO_2$/y over 180 and 500 km, respectively (ZEP, 2010). For similar cases, our study estimated the costs on 12 and 13 €/tCO_2, respectively. The main reason for the slightly lower costs is the higher electricity costs of 110 €/MW h, which are used in the study of ZEP (ZEP, 2010). Implementing higher electricity costs increases the levelized costs to 14 €/tCO_2 and 15 €/tCO_2. Furthermore, ZEP estimated the levelized cost for 20 $MtCO_2$/y over 180 and 500 km at 10 €/tCO_2 and 11 €/tCO_2, respectively (ZEP, 2010). Our study estimated the costs on 8.0 and 9.6 €/tCO_2, respectively. After correction of the electricity costs, the levelized costs increase to 9.9 €/tCO_2 and 12 €/tCO_2, respectively. The levelized costs estimated by our study are, after correction of the electricity price, within a range of 10% of the costs of ZEP.

In conclusion, the levelized cost for ship transport quoted in literature are comparable with the cost figures presented in this study.

7.2. Limitations of this study

Three main limitations have been identified. Firstly, the distance between the capture location and the storage reservoir was assumed to be similar for both transportation modes. However, this may not be the case, due to geographical circumstances. Furthermore, if the CO_2 source is not located in the harbor region, the pipeline may bypass the harbor resulting in a shorter distance to the storage reservoir. Nevertheless, the conclusions that ship transport is only interesting for relatively small volumes over long distances, for short project durations (like demonstration projects) or for small emitters with no access to suitable CO_2 infrastructure, do not change.

Secondly, with the included uncertainties, the options to abandon the project and switching off the capture unit temporarily were very valuable for the 1.0 $MtCO_2$/y and 2.5 $MtCO_2$/y shipping cases. However, the perceived value of these options seems not only depend on the rate of uncertainty, but also on the attitude and culture of the investors (Sanders et al., 2013). For some type of investors, the value of switching temporarily off the capture unit and abandoning the ships was not recognized and even considered as a lack of commitment to the investment decision (Sanders et al., 2013). Investors asked for (more) guarantees to avoid downtime

[10] The submerged turret loading system and liquefaction would add 0.6 €/tCO_2 and 4.0 €/tCO_2 to the levelized costs, respectively.

[11] The variable costs for pumping to 20 MPa instead to 7 MPa are (0.44 kW h/tCO_2/MPa × (20 MPa−7 MPa) × 0.220 kg_{fuel}/kW h × 0.790 €/kg_{fuel}=) 0.99 €/tCO_2. The required design pumping capacity is ((13 MPa−7 MPa)/75%/850 kg/m^3 × (15.45 × 10^6 tCO_2/y/(365 × 24 × 3.6 × 0.85)=) 5.4 MW and 12 MW for an outlet pressure of 7 MPa and 20 MPa, respectively. The CAPEX would be (74.3 × (4400/3)$^{0.58}$ × $3^{0.9}$=) 15 M€ and 30 M€, respectively. This leads to an estimated levelized cost difference of (0.99 €/tCO_2 + (30 M€ − 15 M€) × 11.02%/13.1 $MtCO_2$/y=) 1.1 €/tCO_2.

and the possibility of abandonment. Hence, they tried to minimize the need for flexibility rather than maximize the flexibility within the project (Sanders et al., 2013). This will give pipelines an additional advantage compared to ship transport.

Lastly, this study was limited to the techno-economic aspects of ship and pipeline transport. However, also other considerations may play a role in the decision to invest in pipeline or ship transport, like typical weather conditions, space in the harbor, opportunity to use existing infrastructure, experience with one of the transportation modes, regulation, environmental impacts, etc. These considerations are very case specific and were, therefore, outside the scope of this study.

8. Conclusions and further research recommendations

8.1. Conclusions

The aim of this study was to investigate the value of flexibility for CO_2 ship and pipeline transport and assess whether this flexibility value influences the investment decision between them.

The results of the NPV approach show that pipelines are the preferred transportation mode for design capacities of 2.5 $MtCO_2$/y or 10 $MtCO_2$/y over 250 km, while ships are preferred for 1.0 $MtCO_2$/y over 250 km, and 1.0 or 2.5 $MtCO_2$/y over 500 km. Ships are also the preferred transportation mode for a design capacity of 10 $MtCO_2$/y over 500 km, if the limited storage capacity of 100 Mt is taken into account. All these projects are not profitable with an initial CO_2 price of 35 €/t, which increase with 3% per year. The profitability of the CCS project is mainly influenced by the CO_2 price. For instance, with a distance of 250 km and capacity of 10 $MtCO_2$/y, the initial CO_2 price has to increase to 42 €/t for the pipeline and 38 €/t for the ship configuration to make the project profitable.

The results of the ROA show that the option to switch to another storage reservoir is very valuable for the 10 $MtCO_2$/y cases, with option values ranging from 1012 to 1177 M€. Consequently, it makes the project values positive for both transportation modes for the 10 $MtCO_2$/y cases over a distance of 250 and 500 km. However, the option value for the 10 $MtCO_2$/y pipeline configuration is about 10% higher than for the shipping configuration, despite the much higher costs of making a switch to another reservoir. The reason is the higher (estimated) revenues after a switch is made. For the 2.5 $MtCO_2$/y case, the option value is twice as high for the ship than for the pipeline configuration, because there is only limited time to earn back the switching costs. However, this option value is only a few thousand Euros. Overall, the value of switching to another reservoir is higher for the pipeline configuration if there is a long time that the CO_2 can be stored, while the option is more valuable for the ship configuration if the time left to store CO_2 is only limited.

The options to switch off the capture unit temporarily and abandon the CCS project are more often applied for the ship than for the pipeline configurations. The probability of switching off the capture unit is about 3% and 7–10% for the pipeline and ship configurations, respectively. Consequently, the value of the switching off option is about five times as high for the ship compared to the pipeline configurations for the 1.0 and 2.5 $MtCO_2$/y cases. For the same design capacities, the option to abandon the project is about two to four times as valuable for the ship than for the pipeline configuration. The main reason being the higher variable OPEX and the higher residual value of the former. In none of the analyzed cases, the project value turned positive or the preferred transportation mode changed by including the option to switch off the capture unit temporarily or abandon the project.

With all three options combined, the option to temporarily switch off the capture unit has a strong interaction effect with abandoning the project and is consequently hardly applied in any of the analyzed cases. The option value of the 1.0 and 2.5 $MtCO_2$/y configurations is mainly driven by the abandoning option, while the switching field option is the main driver of the option value of the 10 $MtCO_2$/y configurations. For the 10 $MtCO_2$/y configurations, the option value of all options together range from 1050 to 1183 M€ and makes the project value positive for both considered transportation modes and both distances. However, the project values remain higher for the pipeline than for the shipping configurations. For design capacities of 1.0 and 2.5 $MtCO_2$/y, the combined option values are about 10–14 M€ for the pipeline and 27–34 M€ for the shipping configurations. However, the overall project value remains negative and the preferred transportation mode does not change for these design capacities.

Overall, this analysis showed that both 10 $MtCO_2$/y cases are profitable with the ROA, while they were not profitable with the NPV approach with an initial CO_2 price of 35 €/tCO_2, which increase with 3% per year. Hence, incorporating the value of the most relevant flexibility options should be done, when the standard NPV approach gives negative results to avoid that profitable projects are not conducted.

Although the flexibility that ships offer in comparison with pipelines is often mentioned in literature, the results of this study show that the value of flexibility did not change the investment decision from pipeline to ship transport, at least for the considered options to abandon the project, switch off the capture unit temporarily and switch to another storage reservoir. For the last two options, this is mainly caused by the 50% higher variable OPEX of the ship in comparison with the pipeline configurations. For the abandoning option, it should be realized that the other components of the shipping chain (like the liquefaction unit, onshore and offshore temporary storage) represent about 70–80% of the costs and these components are (almost) as inflexible as pipelines.

In general, the findings of this research indicate that the use of ROA provides insights into understanding whether more expensive alternatives (with presumed flexibility) are worth it. This is not only valid for infrastructure related investment decisions but also for other type of investment decisions, such as comparing different type of machines or equipment, or even renting versus buying software etc. The method described in this study, based on the least-square Monte Carlo method, is very suitable for comparing alternatives which have different flexibility characteristics.

8.2. Further research recommendations

This study is a first attempt to incorporate the value of flexibility by comparing CO_2 pipeline and ship transport. Several research recommendations can be made:

- Further investigation is needed to assess the necessity for offshore temporary storage, the most effective way of attaching the ship to the injection well, and the conditioning requirements. In this study, it is assumed that a floating vessel is needed to temporarily store the CO_2 offshore, a submerged turret loading system and a spread mooring system are used for coupling the ship to the injection wells, and conditioning is needed to increase the temperature to 0 °C and pressure to 7 MPa. These three parts represent about 1/3 of the total costs for transporting annually 2.5 $MtCO_2$ with ships.
- Several cost relations for the shipping case are based on only one or two data points. In other cases, large cost ranges are mentioned in literature. Especially the costs of liquefaction of pre-compressed CO_2, floating vessel, and offloading system require validation, because these components are not only based on

relatively limited costs data but also have a relatively large share in the overall levelized costs for ship transport.

- Liquefaction based on a multi-stage compression configuration is 44% more efficient than a one-stage configuration (Yoo et al., 2013b). In this study, the costs are based on a one-stage configuration, because CAPEX values were not available for a multi-stage liquefaction configuration. Hence, it should be investigated whether, and if so how much, the levelized costs could be reduced if liquefaction would be based on a multi-stage instead of a one-stage compressor configuration.
- Not much information is (publically) available on the abandonment costs of a power plant, an offshore pipeline, a liquefaction plant, etc. More insights have to be gained into the ABEX of these components.
- The option to make a switch to another storage reservoir was very cost-effective for large mass flows. However, the costs of realizing such a switch should be validated for ship as well as for pipeline transport.
- The option to expand the amount of CO_2 transported over time and the option to postpone the investment decision are also interesting options for CO_2 ship and pipeline transport. It is recommended to also investigate the value of these options in the future.

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